Ch- Atomic Structure

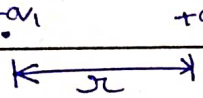
Properties of Charge :-

(1) $q = ne$, $n = 1, 2, 3, \cdots$ (integer)

$e^- \rightarrow$ Charge one $e^- = 1.602 \times 10^{-19}$ C (SI)

$= 4.8 \times 10^{-10}$ (esu)

(2) Coulomb's law



$F = \frac{K q_1 q_2}{D \times r^2}$ [Inverse square law]

D → Dielectric Constt.

D = 1 for air, D = 80 for water

$K = \frac{1}{4\pi\epsilon} = 9 \times 10^9$ in SI Unit

$= 1$ in CGS Unit

where ϵ_0 → Permitivity of free space

→ For like charges - Repulsive force

→ For unlike charges - Attraction force

Potential Energy (U)

$U = \frac{K q_1 q_2}{r} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$

→ For like charges, U → +ve (Repulsive)

→ For unlike charges, U → -ve (Attractive)

(3) Electric force on a charged particle due to Electric field

$F_e = qE$

(4) Magnetic force on charged particle due to Magnetic field

$F_m = qvB$

$F \perp v \Rightarrow$ Circular Motion

Imp

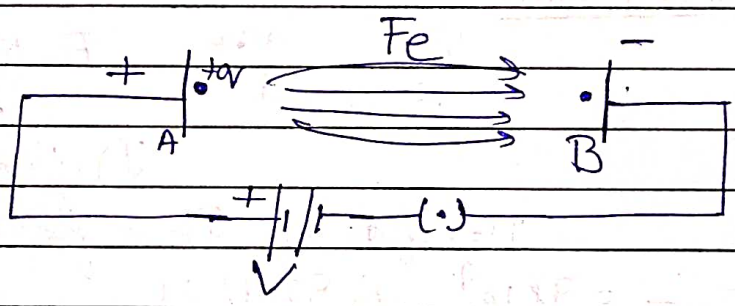
(5) When a potential diff of V volt is applied on a charged particle at rest, then Kinetic Energy acquired by charged particle.

$$KE = q \times V$$

Or

$$\boxed{\frac{1}{2} m v^2 = q \times V}$$

→ Proof



Applying law of conservation of Energy

$$(TE)_A = (TE)_B$$

$$(PE)_A + (KE)_A = (PE)_B + (KE)_B$$

$$q \times V + 0 = 0 + \frac{1}{2} m v^2$$

$$\Rightarrow \boxed{KE = \frac{1}{2} m v^2 = q \times V}$$

m = mass of charged particle
v = velocity acquired by charged particle
V → Pd in volt.

Unit of Energy of charged Particle :
SI Unit : Joule

→ Another Useful unit is (eV)

(1eV) → It is kinetic energy acquired by an electron initially at rest under an accelerating PD of 1 volt.

$KE = q \times V$

$= 1e \times 1 \text{ volt}$

$= 1eV = 1 \times 1.6 \times 10^{-19} C \times V = 1.6 \times 10^{-19} J$

$\boxed{1eV = 1.6 \times 10^{-19} \text{ Joule}}$

Q Find KE of an electron at rest when a P.d of 75 volt is applied across it.

Ans $KE = q \times V$

$= 1e \times 75V$

$= 75eV$

$= 75 \times 1.6 \times 10^{-19}$ Joule

Q Find final KE of an alpha particle (α-particle) initially at rest and accelerated by a Pd of 100 Volt?

Ans $KE = q \times V = (2e) \times 100 \text{ Volt}$

$= (200eV)$ Ans $= 200 \times 1.6 \times 10^{-19} J = 3.2 \times 10^{-17} J$

★ Note :→ α- Particle (nucleus of Helium Atom) i.e He^{2+}, charge $= +2e$

→ Proton or H^+, charge $= +1e$

→ Electron, charge $= -1e$

→ Deuteron (D^+) charge $= +1e$

→ Triton (T^+), charge $= +1e$

→ Neutron, charge $= 0$

Q An α-particle having KE of 200eV. It is further accelerated by a P.d of 50 Volt. Find final KE of particle.

Ans $(KE)_f = 200eV + qXV$

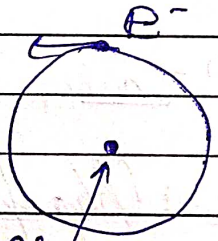
$= 200eV + 2e \times 50$

$= 300eV$

Q Both α-particle & proton are moving with initial KE of 100eV. Both are decelerated by a PD of 20 Volt. Find ratio of KE of α and proton finally.

Ans $\left\{\frac{(KE)_\alpha}{(KE)_p}\right\}_f = \frac{100 - 2e \times 20}{100 - 1e \times 20} = \frac{60}{80} = \frac{3}{4}$ Ans

Q Write the expression of Potential energy b/w electron and nuclear charge in Li^{2+}



Charge on nucleus $= +Ze$

Z = Atomic no.

= No of p^+

For Li, $Z = 3$

$PE, V = \frac{Kq_1q_2}{r} = \frac{1}{4\pi\epsilon_0} \times \frac{(+3e)(-1e)}{r}$

$= -\frac{1}{4\pi\epsilon_0} \times \frac{3e^2}{r}$

Specific Charge (q/m):- It is charge and mass ratio of a ~~Proton~~ Particle.

Q Compare specific charge magnitude (i.e magnitude of $\frac{q}{m}$) for electron, proton, α-particle, neutron.

Ans For solving Problems

$m_p \cong m_n = 1840 \times m_e$

$m_\alpha = 4 \times m_p$

	e	p	α	n
$\frac{q}{m}$:-	$\frac{1e}{\left(\frac{m_p}{1840}\right)}$	$\frac{1e}{m_p}$	$\frac{2e}{4m_p}$	$\frac{0}{m_p}$
	$= \frac{1840 \times e}{m_p}$	$\frac{1 \cdot e}{m_p}$	$\frac{1}{2}\frac{e}{m_p}$	0

$\Rightarrow$ $n < \alpha < p < e$ (increasing $\frac{q}{m}$ of particles)

Q Find ratio of specific charge or $\frac{q}{m}$ of deuteron and α-particle.

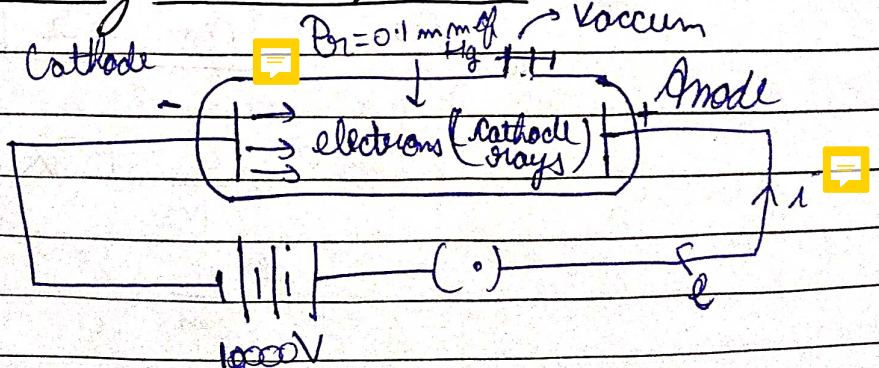
Ans $D^+ \begin{pmatrix} 1p \\ 1n \end{pmatrix}$ or $He^{2+} \begin{pmatrix} 2p \\ 2n \end{pmatrix}$

$\left(\frac{q}{m}\right)_{D^+} = \frac{1e}{2m_p}$, $\left(\frac{q}{m}\right)_\alpha = \frac{2e}{4m_p}$

$\frac{\left(\frac{q}{m}\right)_{D^+}}{\left(\frac{q}{m}\right)_\alpha} = \frac{1}{2} \times \frac{2}{1} = 1:1$

$\Rightarrow$ Discovery of Electron: (By Thomson)

Discharge Tube Experiment:



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→ Specific charge ($\frac{q}{m}$) of cathode rays particles is independent of nature of

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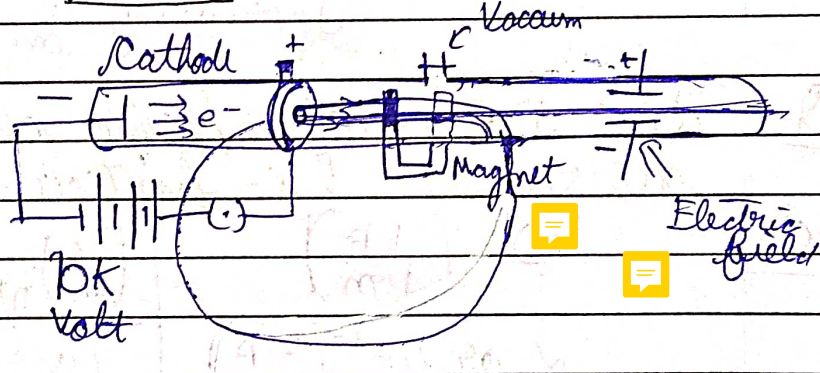
→ Specific Charge ($\frac{q}{m}$) of cathode rays particles is independent of nature of gas filled in discharge and also independent of nature of material of cathode / anode.

→ Thomson concluded that e^- (particles of cathode rays) is a fundamental particle of every matter.

⇒ Determination of $\frac{q}{m}$ of e^- by Thomson :-

(can be skipped)



1. $qvB = \frac{mv^2}{r}$

$$\boxed{V = \frac{qBr}{m}} \quad \text{—①}$$

2. $qvB = qxE$

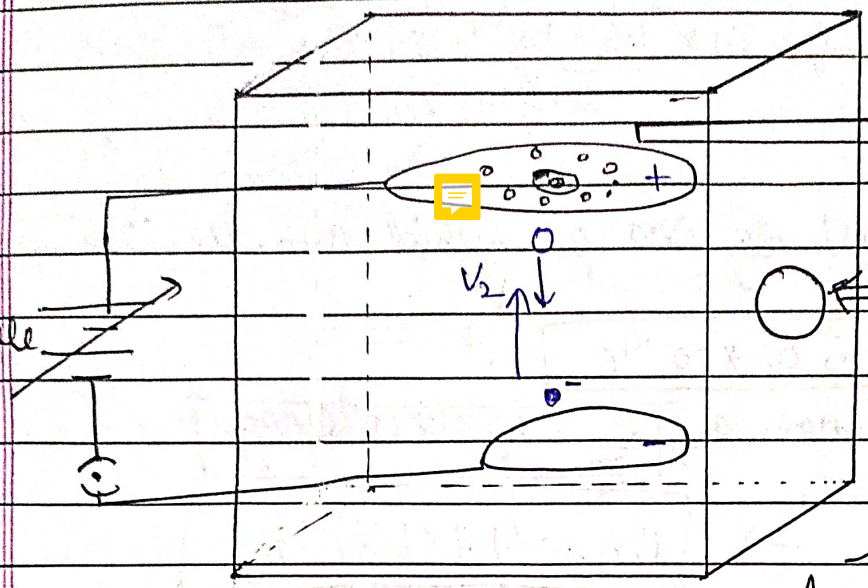
$$\boxed{V = \frac{E}{B}} \quad \text{—②}$$

$$\frac{E}{B} = \frac{qBr}{m} \Rightarrow \boxed{\frac{q}{m} = \frac{E}{B^2 r}}$$

For e^-, $\frac{q}{m} = 1.76 \times 10^{11}$ C/Kg

or $\frac{e}{m}$

Millikan's Oil Drop Experiment (Skippable)



→ Air inside the vessel gets ionised by X-rays O_2^+, N_2^+, O_2^{2+}

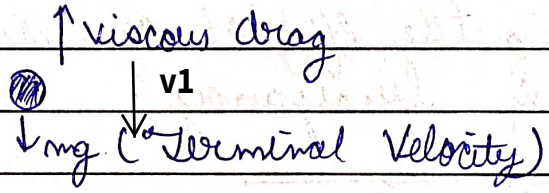
Atomizer

X-rays

→ e^- are captured by oil drops and become -vely charged.

Using Stoke's law

Step-1:

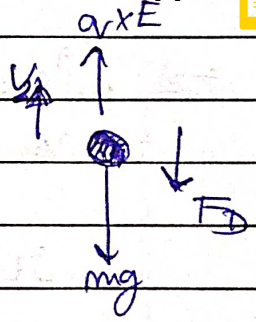


$$F_D = 6\pi\eta r v_1 = mg$$

(Viscous Drag)

v_1 is experimentally measured by microscope

Step-2: P.D is applied



$$qE = mg + 6\pi\eta r v_2$$

$$\Rightarrow q = \frac{6\pi\eta r(v_1+v_2)}{E}$$

→ Radius of oil drop

$$6\pi\eta r v_1 = d_{oil} \times \frac{4}{3}\pi r^3 g$$

$$r^2 = \frac{9\eta v_1}{2 d_{oil} g}$$

$$\Rightarrow r = \sqrt{\frac{9\eta v_1}{2 d_{oil} g}}$$

⇒ From here we can find out charge q.

→ Millikan observed that

$q = 1.6 \times 10^{-19}$ C, 3.2×10^{-19} C, 4.8×10^{-19} C

$\Rightarrow q = n \times (1.6 \times 10^{-19})$

where $n = 1, 2, 3, 4$

☆ ⇒ Minimum unit of charge which can be transferred is 1.6×10^{-19} C.

$\therefore$ $\boxed{e = 1.6021 \times 10^{-19} C}$

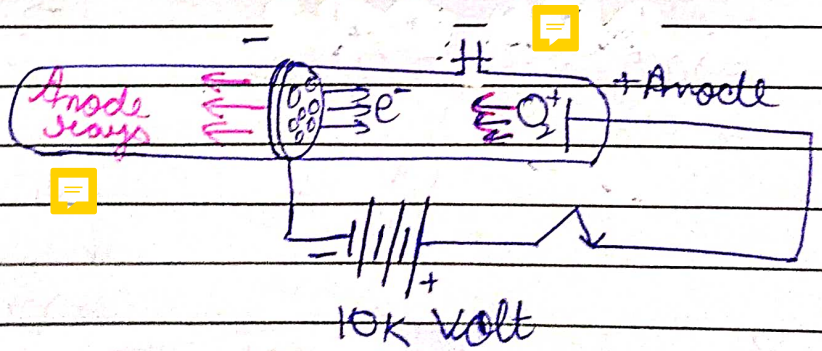
⇒ Acc to Thomson, $\boxed{\frac{e}{m} = 1.76 \times 10^{11} C/kg}$

$\Rightarrow$ $\boxed{m_e = 9.1 \times 10^{-31} kg}$

⇒ Considering the effect of medium (air), or considering buoyancy.

$$r = \sqrt{\frac{9 \eta v_t}{2(d_{oil} - d_{air})g}}$$

⇒ Discovery Of Proton :-



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→ $\frac{q}{m}$ of anode rays particles depend on nature of gas filled in discharge tube. Actually particles of anode rays are ionised gas particles.

→ If atomic hydrogen is filled in discharge tube then the particles of anode rays are called **Protons**

Discovery of Neutron :-
(By Chadwick)

Electron – Thomson
Proton – Goldstein
Neutron – Chadwick

→ Chadwick bombarded α - particles on Be foil and observed that highly penetrating rays were produced

→ Also these rays were not deflected in any electric or magnetic field, indicates that particles are neutral, called these particles Neutrons.

$$_{4}Be^{9} + {}_{2}He^{4} \longrightarrow {}_{6}C^{12} + {}_{0}n^{1}$$

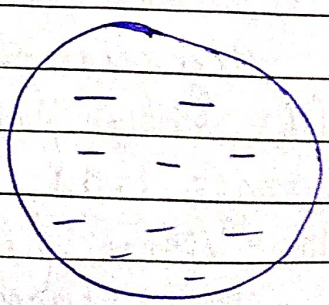
(He^4: produced by a radioactive source; n^1: Neutron)

* $m_n \cong m_p$ (for numerical questions)

Various Atomic Models :

① **Thomson's atomic model :**
(Plum Pudding Model)

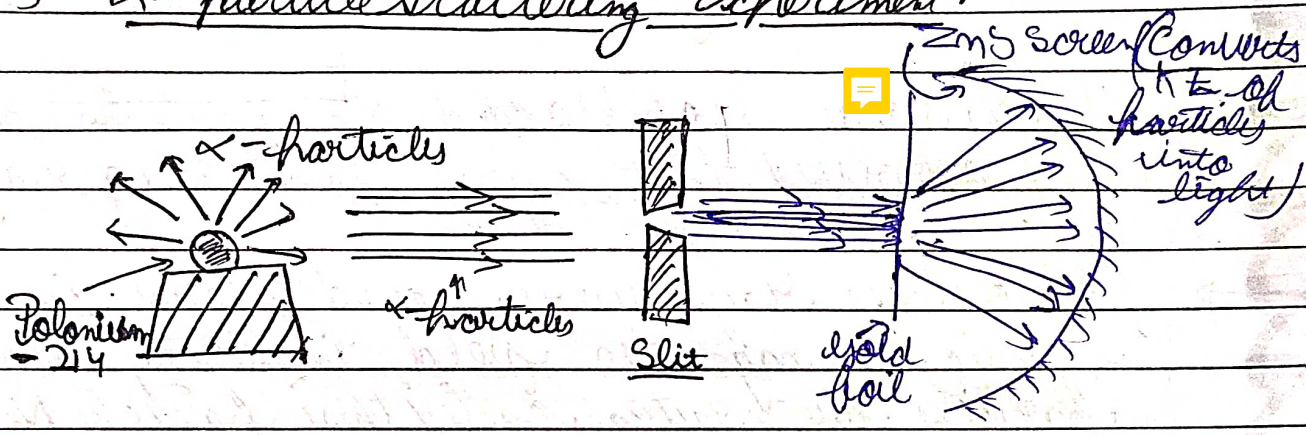
→ Electrons are imbedded in the sphere of size of atom where the +ve charge is uniformly spread and atom is electrically neutral.

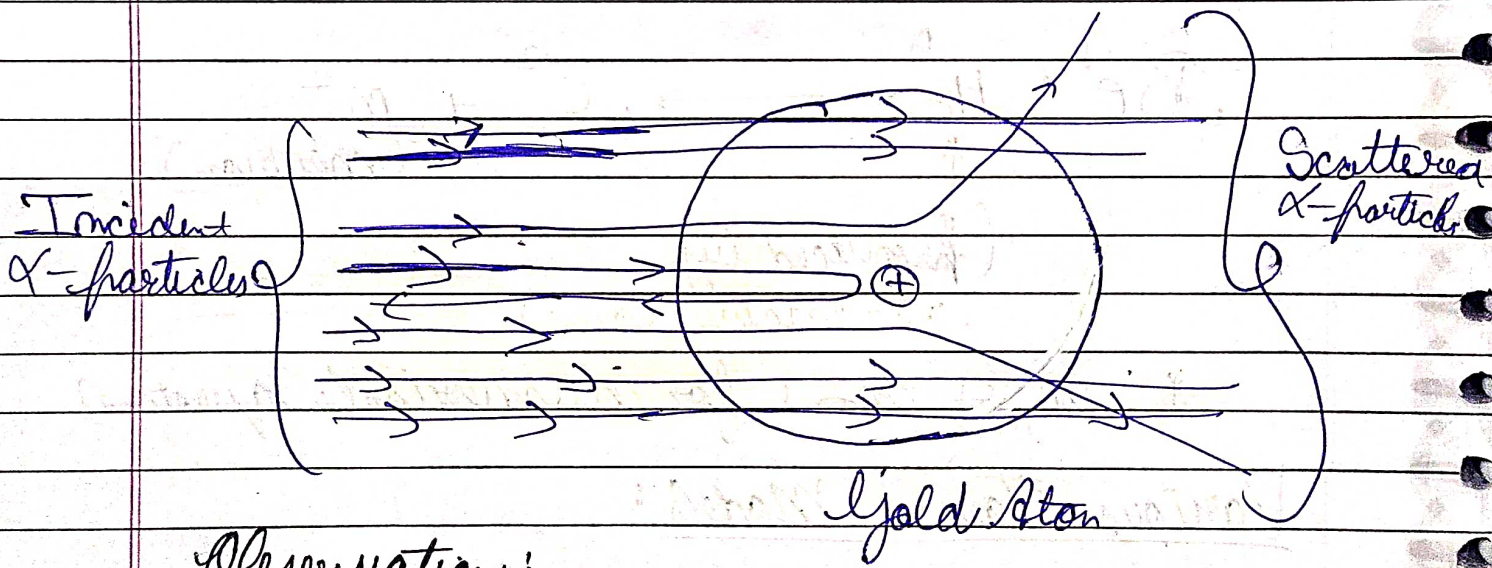


→ This model failed as it was unable to explain ~~α-p~~ Rutherford's α-particle scattering experiment.

(2) Rutherford's Nuclear Model :-

⇒ α-particle scattering Experiment:





Observations:

(i) Most of the α particles went straight undeflected

(ii) Only few got deflected away to an angle as large as 90°.

(iii) Very few particles (nearly 1 in 20000) got recoiled or returned

Conclusions :- (i) Most of the space of atom is empty.

(ii) There must be a dense +ve charged centre which could deflect +vely charged α-particles away from it.

⇒ Rutherford called this +vely charged center as nucleus.

Rutherford's Nuclear Model :-

(i) Most of the space of the atom is empty because all electronic charge is spread into the large spherical volume of size of atom i.e. charge density of electrons is very low.

(ii) All the charge and almost entire mass of atom is ~~considered~~ concentrated into nucleus (extremely small size ~~as~~ relative to size of atom)

(iii) Since electrons are revolving around the nucleus the force of attraction b/w electron & nucleus is balanced by centrifugal force.

(iv) Size of atom is of the order of 10^{-10} m whereas that of nucleus is of order of 10^{-15} m.

Determination of size of atom:

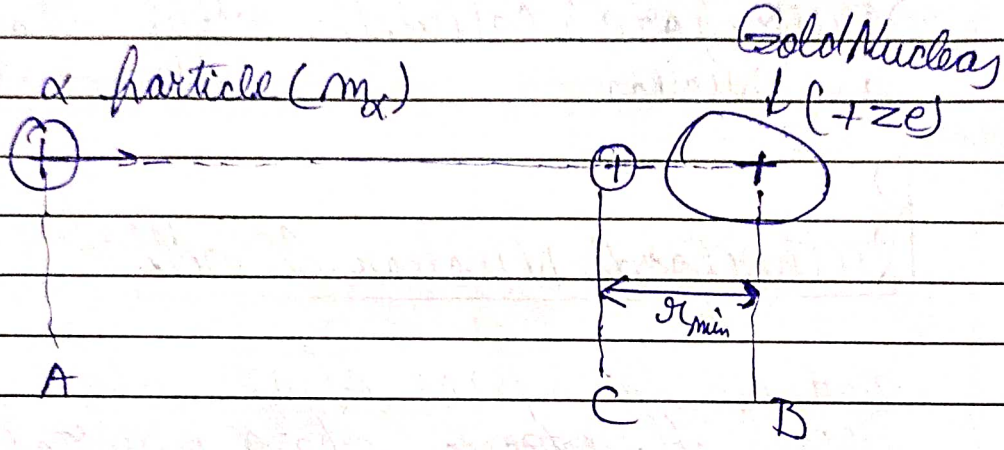
$d_{Al(s)} = 2.7$ gm/ml, Molar mass of Al = 27 g/ml

$V_{molar} = \frac{27}{2.7}$ cm^3 $\Rightarrow 6 \times 10^{23} \times \frac{4}{2} \pi r^3 = 10$

$\Rightarrow r^3 = \frac{100}{8\pi} \times 10^{-24} \Rightarrow r \approx 10^{-8}$ cm $= 10^{-10}$ m

Determination of Approximate size of Nucleas:

→ Consider the head on collision b/w α-particle and gold nucleas (when α-particle is recoiled)



r_{min} → Distance of closest approach

Applying law of conservation of Energy -

$$(TE)_A = (T.E)_C$$

$$(KE)_A + (PE)_A = (KE)_B + (PE)_B$$

$$\frac{1}{2} m_\alpha v_\alpha^2 + 0 = 0 + \frac{K(+2e)(+Ze)}{r_{min}}$$

$$\Rightarrow \boxed{r_{min} = \frac{2KZe^2}{\left(\frac{1}{2} m_\alpha v_\alpha^2\right)}}$$

$$\Rightarrow \boxed{r_{min} = \frac{2KZe^2}{(KE_i)_\alpha}}$$

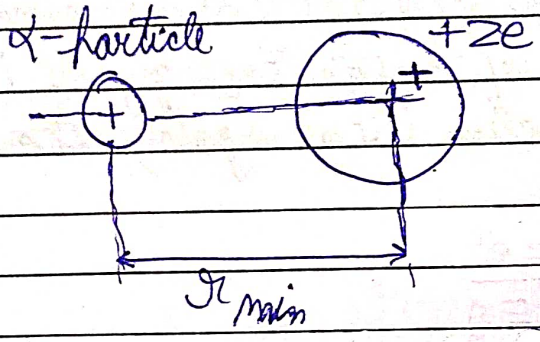
↑ Distance of closest approach (Approximate size of nucleas)

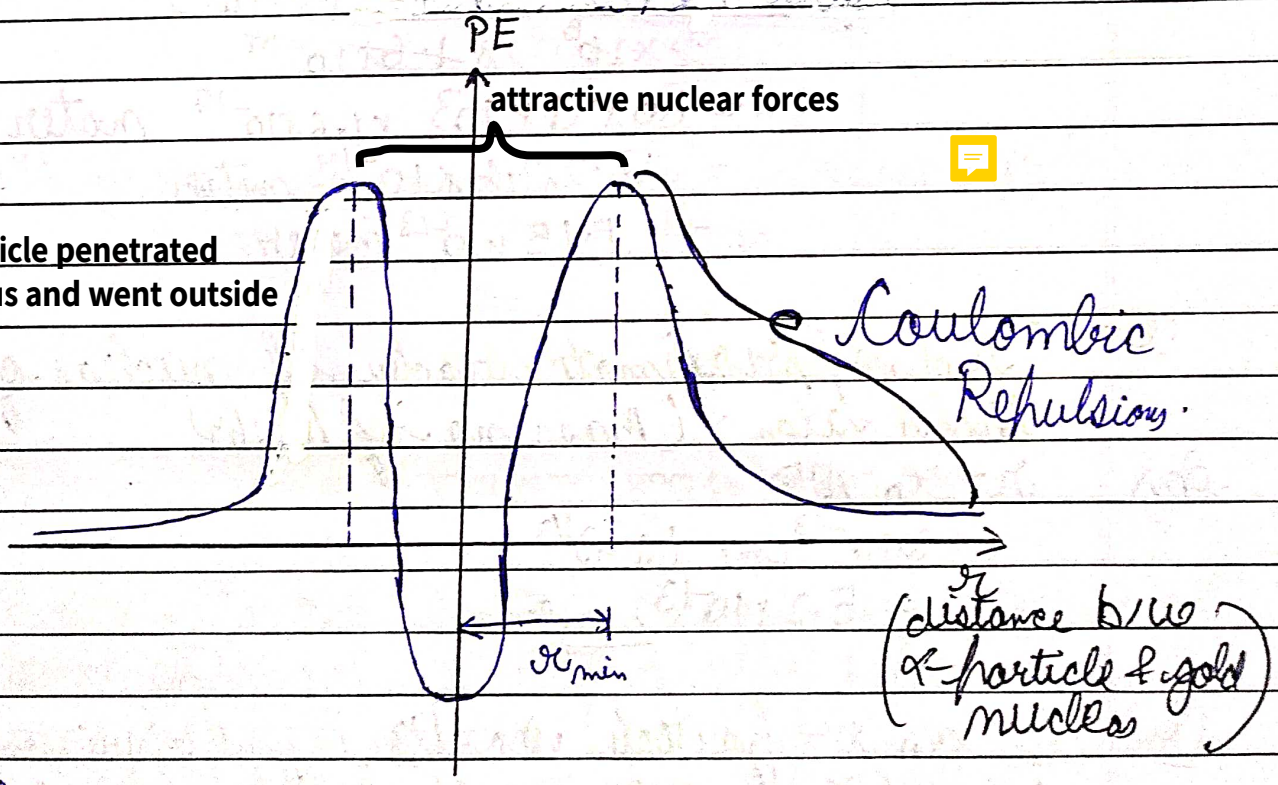
$\Rightarrow r_{min} \propto Z$

$\propto \frac{1}{KE}$

or $\propto \frac{1}{V_\alpha^2}$

$\Rightarrow$





(Postulate)

(V) Approximate size of nucleus -

$r \propto A^{1/3}$

or $r = r_0 A^{1/3}$

r = radius of nucleus

where r_0 = a constant
$= (1.3 \text{ to } 1.4) \times 10^{-13}$ cm

A = Mass no.

(vi) No. of α-particle at an angle θ,

$$N_\theta \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$

Q Calculate the approximate size of nucleus (i.e distance of closest approach when α-particle of Kinetic Energy of MeV collides the gold nucleus (For Gold, Atomic no = 79)

Ans
$$r_{min} = \frac{2KZe^2}{(KE_\alpha)}$$
$$= \frac{2 \times 79 \times 9 \times 10^9 \times (1.6 \times 10^{-19})^2}{2 \times 10^6 \times 1.6 \times 10^{-19}}$$
$$\simeq 80 \times 9 \times 10^3 \times 1.6 \times 10^{-19} \text{ meter}$$
$$= 7.2 \times 1.6 \times 10^{-14} \text{ meter}$$
$$= 1.15 \times 10^{-13} \text{ meter}$$

Q Find approximate radius of nucleus of Nickel atom. (Mass no. of Ni: 64)

Ans
$$r = r_0 A^{1/3}$$
$$= 1.3 \times 10^{-13} \text{ cm} \times (64)^{1/3}$$
$$= 5.2 \times 10^{-13} \text{ cm Ans}$$

Q In an α-particle scattering experiment, No of α-particles scattered at an angle of 90° are 280. Find no of α-particles scattered at 180°.

Ans
$$N_\theta \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$
$$\frac{N_{\theta_2}}{N_{\theta_1}} = \left\{\frac{\sin\left(\frac{\theta_1}{2}\right)}{\sin\left(\frac{\theta_2}{2}\right)}\right\}^4 = \frac{\left(\frac{1}{\sqrt{2}}\right)^4}{1^4} = \frac{1}{4}$$

$\Rightarrow N_{180^\circ} = \frac{200}{4} = 50$ α-particle.

Q In an α-particle scattering experiment if distance of closest approach is r. If velocity of α-particle is increased by 50%, find % change in distance of closest approach.

Ans $r_{min} = \boxed{\frac{2KZe^2}{\frac{1}{2}mv_\alpha^2}} \Rightarrow r_{min} \propto \frac{1}{v_\alpha^2}$

$$\frac{r_2}{r_1} = \left(\frac{v_1}{v_2}\right)^2$$

$$\frac{r_2}{r_1} = \left(\frac{v}{1.5v}\right)^2 = \frac{1}{2.25}$$

$$\boxed{r_2 = \frac{r_1}{2.25}}$$

$\Rightarrow$ % decrease in distance of closest approach

$$= \frac{r - r_e\,|2.25}{r} \times 100$$

$$= \frac{1.25}{2.25} \times 100$$

$$= \frac{5}{9} \times 100\% = \frac{500\%}{9} = 55.5\%$$

Q In an α-particle scattering experiment, distance of closest approach is r. If another experiment is conducted with another metal foil of atomic number is 1.5 times and KE of α-particles is doubled. Find % change in distance of closest approach.

Ans $r_{min} = \frac{2KZe^2}{(KE)} \Rightarrow \frac{r_2}{r_1} = \frac{Z_2}{Z_1} \times \frac{KE_1}{KE_2}$

$$\frac{r_2}{r} = 1.5 \times \frac{1}{2}$$

$$r_2 = \frac{3}{4} r = 0.75\, r$$

% decrease $= \frac{r - 0.75r}{r} \times 100 = 25\%$ Ans

Failure of Rutherford's Model :-

(1) According to Maxwell's wave theory if any charged particle is moving under acceleration, then it must lose energy since is revolving around nucleus with radial acceleration $\left(\frac{v^2}{r}\right)$, it should lose energy and should fall inside the nucleus.

→ However it doesn't happen. Why?

Rutherford could not explain about the stability of the nucleus.

(2) Rutherford couldn't explain the line spectrum of atomic hydrogen

Some Important terms:

(1) ${}^{A}_{Z}X$ or ${}^{}_{Z}X^{A}$ ← Mass number = Sum of protons & Neutrons (or total no of nucleons)

Z ↑ Atomic no = No of protons = No of electrons

(2) **Isotope** :- Atoms of same element which have same atomic no. but different mass number.

eg $_1H^1$ (Protium), $_1H^2$ (D) Deuterium, $_1H^3$ (T) Tritium (Radioactive)

eg $_{17}Cl^{35}$, $_{17}Cl^{37}$ eg $_6C^{12}$, $_6C^{14}$ (Radioactive)

(3) **Isobars**:- Atoms of different elements which have same mass number but different atomic number.

eg $_2He^3$, $_1H^3$

Eg $_{18}Ar^{40}$, $_{19}K^{40}$, $_{20}Ca^{40}$

(4) **Isotones** :- which have same number of neutrons (i.e same A-Z value)

Eg $_2He^4$, $_1H^3$

$(A-Z) = 4-2$, $3-1$

$= 2 : 2$

(5) **Isoelectronic Species**:- Same total number of electrons.

eg N_2O, CO_2

$22e^-$ $22e^-$

eg N^{3-} O^{2-}, F^-, Ne, Na^+, Mg^{2+}, Al^{3+}

isoelectronic species

(6) **Isodiaphers :-** which have difference of no of neutrons (A-Z) and protons (Z) i.e Same (A-Z) value

Eg $_{92}U^{235}$ & $_{90}Th^{231}$

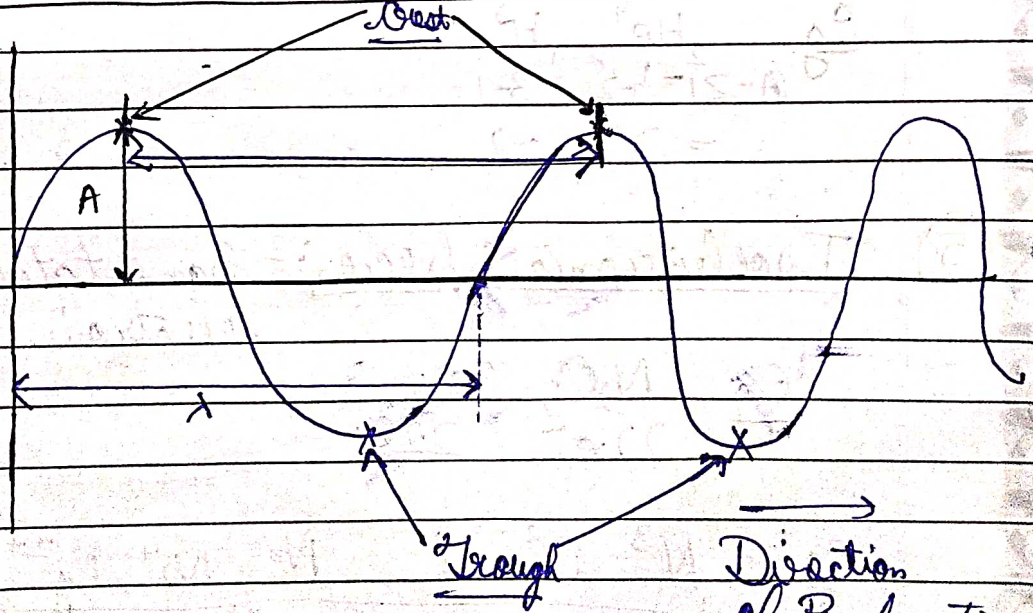
(A-Z) = 235-184	231-180
= 51	= 51

Wave :- It is a means of transfer of energy from one place to another without transfer of molecules of medium.

→ For Electromagnetic wave's propagation Medium is not needed.

→ In wave motion, oscillations propagate or move and transfers energy.

Some terms used in wave motion :-



Sinusoidal Wave (Sin θ vs θ)

① **Time Period (T)**: Time taken by wave for one complete oscillation

$$v = \frac{\lambda}{T} \Rightarrow \boxed{T = \frac{\lambda}{v}}$$

② **Wavelength (λ)**: Linear distance travelled by wave in one complete oscillation

or

It is the linear distance b/w two consecutive crests or troughs.

$$v = \frac{\lambda}{T}, \quad T \rightarrow \text{Time Period}$$
$$v \rightarrow \text{Velocity of wave}$$

→ For electromagnetic waves, (e.g. light, X Rays, γ-rays) etc

$$c = \frac{\lambda}{T} \Rightarrow T = \frac{\lambda}{c}$$

Where c = speed of light
$= 3 \times 10^8$ m/s in vaccum

③ **Frequency (ν)**: It is no of oscillations completed (Hertz or sec^{-1}) in 1 sec.

$$T \text{ sec} \text{ ——— } = 1 \text{ oscillation}$$
$$1 \text{ sec} \text{ ——— } = \frac{1}{T} = \nu$$

$$\nu = \frac{1}{T} = \frac{c}{\lambda}$$

or $\boxed{\nu = \frac{c}{\lambda}}$

(4) **Wave Number ($\bar{\nu}$):** It is the no of wavelengths in unit length.

$$\tilde{\nu} = \frac{1}{\lambda}$$

(m^{-1})

(5) **Amplitude (A):** It is height of crest or depth of trough in wave motion.

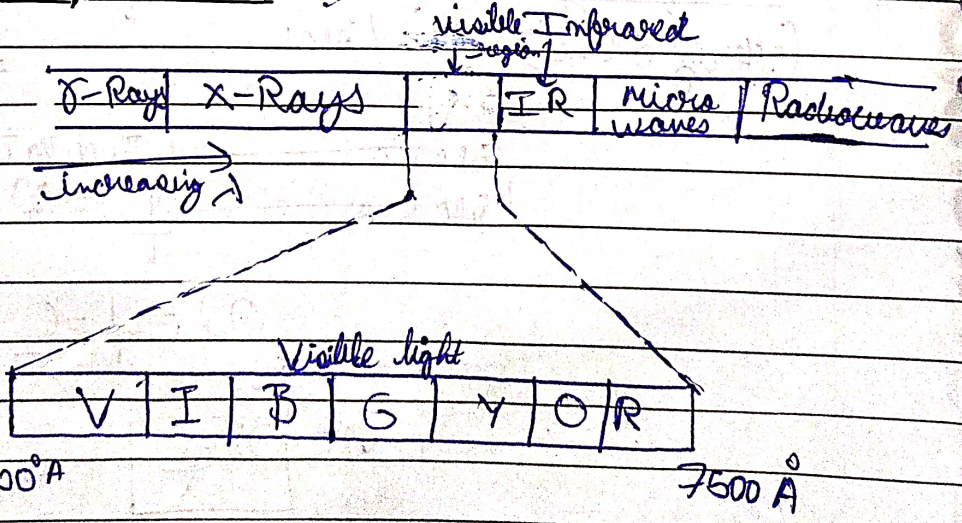
6. A visible light has wavelength equal to 4000 Å. Find its frequency.

Ans $\nu = \frac{c}{\lambda} = \frac{3\times10^{8}}{4000\times10^{-10}} = 0.75\times10^{5} = 7.5\times10^{14}$ Hz

Electro Magnetic Waves: (or EM radiation) EM waves consist of oscillating electric and magnetic field, both field components are ⊥r to each other and also ⊥r to the direction of Propagation of wave.

EM radiations include, radiowaves, microwaves UV, X-rays etc.

EM wave Spectrum :-



→ For all EM Waves, $\lambda = \frac{c}{\nu}$

Q Range of wavelength of visible light is 4000 Å to 7500 Å. Convert this into frequency range in Hz.

Ans 4000 Å to 7500 Å

$\nu = \frac{3\times10^8}{4\times10^{-7}}$ to $\frac{3\times10^8}{7.5\times10^{-7}}$

$= 0.75\times10^{15}$ to 4×10^{14}

$= 7.5\times10^{14}$ Hz to 4×10^{14} Hz Ans

Maxwell's Wave Theory :-

→ All EM radiations move with 3×10^8 m/sec in vaccum.

→ It consists of oscillating electric and magnetic fields, both fields ⊥r to each other and also ⊥r to the direction of propogation of wave.

→ Energy of all EM waves ∝ square of Amplitude
$\propto A^2$

→ Intensity of light = Energy per unit area per unit time
$= \frac{E}{A\times t}$, $J/m^2\cdot sec$

→ Strength of Maxwell's wave theory
↳ Successfully explains
(i) Interference
(ii) Diffraction

→ Weakness → fails to explain
(i) Photo-electric effect
(ii) Blackbody radiations

Planck's Quantum Theory :-
(Particle nature of light)

Atoms or molecules can absorb or emit energy only in discrete quantities (called photon, the quantum of light) and not in continious value / Ammount / manner.

According to Planck,

Energy of each photon $\propto \nu$

$\Rightarrow E = h\nu = h \times \frac{c}{\lambda}$

(h = Planck's Constant $= 6.625 \times 10^{-34}$ Joule-sec)

For large no of photons emitted or absorbed by an object,

$$E = x \times \frac{hc}{\lambda}$$

where x = an integer $= 1, 2, 3, 4, \ldots$

→ This indicates that energy can be emitted or absorbed in integral multiple of $h\nu$.
i.e $1h\nu, 2h\nu, 3h\nu, 5h\nu, \ldots$, etc,
but not fractional multiple of $h\nu$.

$$E = x \times h\nu = x \times \frac{hc}{\lambda}$$

(x = An integer (no of photons))

Dual Nature of light :- Maxwell's wave theory explains Interference and Difraction whereas Planck's quantum theory successfully explains Black Body radiation and Photoelectric effect. None of the theory explains all experimental facts. So light is said to possess dual nature. (Particle as well as wave nature)

Q. A 60 watt bulb emits a monochromatic light of wavelength 6620Å. Find no of photons that are emitted in 1 minute.

Ans Power = $\frac{\text{Energy}}{\text{Time}}$

$\Rightarrow$ Energy = $60 \times 60 = x \times \frac{hc}{\lambda}$

$\Rightarrow \frac{60 \times 60}{20} = x \times \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{6.620 \times 10^{-10-7}}$

$12 \times 10^{2} = x \times 10^{19}$

$x = 1.2 \times 20$ Ans

Note :- $E = \frac{hc}{\lambda}$

↑ Energy of one photon

$hc = 6.625 \times 10^{-34}$ (eV-sec) $\times 3 \times 10^{8} \times 10^{10}$ (Å/sec)

(in eV-Å) 1.6×10^{-19}

$= \frac{6.625 \times 10^{3}}{1.6}$ eV-Å

$hc = 12400$ eV-Å , $E = \frac{12400}{\lambda}$

↑ in eV, ↑ in Å

Q Find Maximum wavelength of light that can break Br-Br bond in Br_2 molecule

$Br_2(g) \longrightarrow 2Br(g)$; $\Delta H_R = 200\ KJ/mol$

1st Approach

$$BE = \frac{hc}{\lambda}$$

$$\Rightarrow \frac{200 \times 10^3}{6 \times 10^{23}} J/molecule = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\frac{1}{3} \times 10^{-18} = \frac{6.62 \times 3 \times 10^{-26}}{\lambda}$$

$$\lambda = 6.62 \times 9 \times 10^{-8}\ m$$

$$= 59.58 \times 10^{-8}\ m\ Ans$$

1eV/atom or Molecule $= 1.6 \times 10^{-19} KJ \times 10^{-3} \times 6 \times 10^{23}$

$= 9.6 \times 10^{1}$

$= 96\ KJ/mol$

$\Rightarrow$ 1eV/atom = 96 KJ/mol

$\simeq$ 23 Kcal/mol

1cal = 4.18 Joule

2nd Approach:- $Br_2 \longrightarrow 2Br$, $\Delta H_R = 200\ KJ/mol$

$$\frac{200}{96} eV/molecule\ of\ Br_2 = \frac{12400}{\lambda}$$

$$\Rightarrow \lambda = \frac{12400 \times 96}{200}^{48}$$

$$= 124 \times 48\ \text{Å}$$

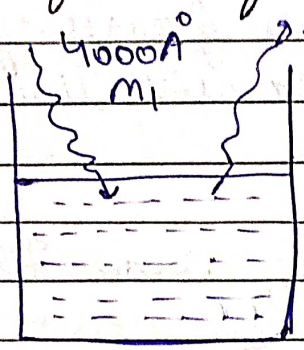
$\Rightarrow$ 1eV/atom = 96 KJ/mol = 23 Kcal/mol

$\Rightarrow \lambda$ (in Å) $= \frac{12400}{E\ (in\ eV)}$

→ light of single wavelength

Q A monochromatic light of wavelength 4000 Å is irradiated on a certain dye. Dye absorb some energy but flores as remaining energy in form of light of 8000 Å. If energy emitted by dye is 50% of the incident energy, find ratio of no. of photons incident and emitted.

Ans



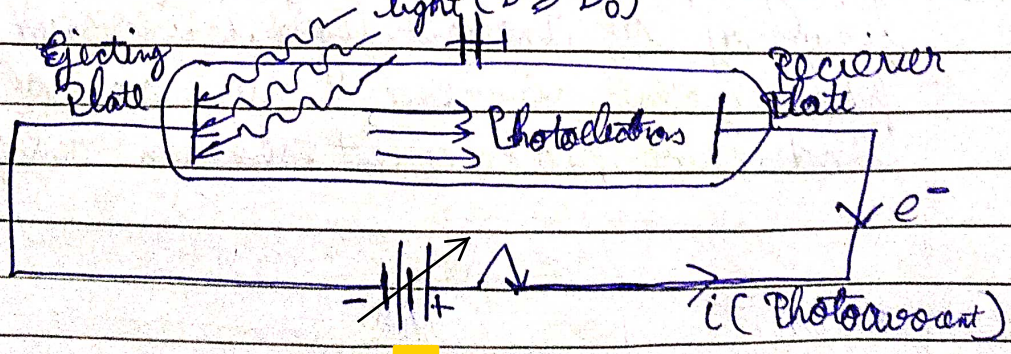
$E_{emitted} = 50\%$ of $E_{incident}$

$$n_2 \times \frac{hc}{8000} = \frac{1}{2} \times n_1 \times \frac{hc}{4000}$$

$n_1 = n_2$

$$\Rightarrow \boxed{\frac{n_1}{n_2} = 1}$$ Ans

Photoelectric Effect :- When a clean metallic surface is irradiated by a monochromatic light of suitable wavelength, photons are ejected from the metal, the phenomenon is called Photoelectric effect and ejected electrons are called Photoelectrons.



Intensity of light :- = Energy of light crossing per unit area per unit time.

$$= \frac{E}{A \times t} = \frac{x \times h\nu}{A \times t} = \frac{x \times hc}{A \times t \times \lambda}$$

→ Photonflux or Photointensity :-

= No of photons of given frequency crossing per unit area per unit time.

$$= \frac{x}{A \times t}$$

* Generally Intensity means Photointensity.

Observations: (Imp)

(1) Photoelectric effect is observed only when frequency of incident light is a certain minimum value, called Threshold Frequency (ν_0)

(2) Keeping frequency of incident light to be same, when photointensity (or intensity) is increased, then no of Photoelectrons and photocurrent linearly increase. However KE of photoelectrons remain unchanged.

(3) Keeping photointensity to be same when frequency of incident light is increased, KE of photoelectrons increase but no. of photoelectrons and hence photocurrent remain unaffected

Note: No of photoelectrons ejected need not be equal to no of photons striking the metal surface.

Einstein's Photoelectric Equation:

$$h\nu = W + (KE)_{max}$$

$$\Rightarrow \boxed{(KE)_{max} = h\nu - W}$$ → Einstein's Photoelectric eq^n

where W = Work function of metal

→ Work function of a metal represents minimum energy required to remove electron from metal

→ Work function of Alkali Metals, particularly K, Rb, Cs are very low.

Li > Na > K > Rb > Cs | Be > Mg > - - - > Ba

decreasing work function | decreasing W

Work function :-

→ The minimum energy needed to remove the electron from a metal surface with zero KE is called work function (W)

$$(KE)_{max} = h\nu - W$$

$$0 = h\nu_0 - W \Rightarrow \boxed{W = h\nu_0}$$

$$\boxed{W = \frac{hc}{\lambda_0}}$$

Threshold Frequency (ν_0) :- The minimum frequency of incident light which can eject a photoelectron of zero KE.

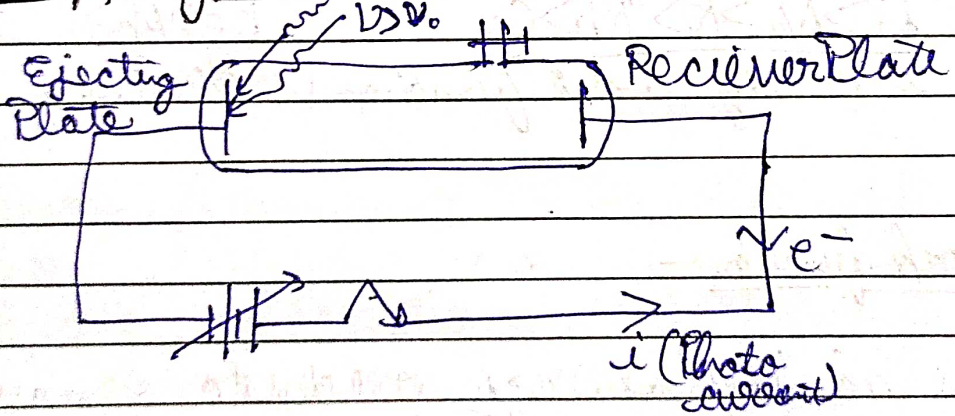
$$0 = h\nu_0 - W \Rightarrow W = h\nu_0 = \frac{hc}{\lambda_0}$$

ν_0 → Threshold frequency; λ_0 → Threshold wavelength.

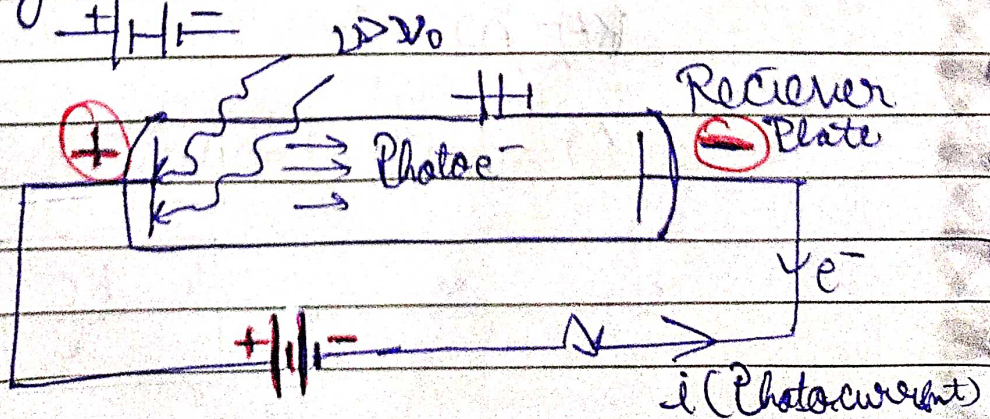
$$\Rightarrow \boxed{(KE)_{max} = h\nu - h\nu_0} \quad \text{where } W = h\nu_0 = \frac{hc}{\lambda_0}$$

Note : KE of photo-e⁻ form a metal may have the value 0 to $(KE)_{max}$.

Stopping Potential (V_0) :



⇒ Now, Battery is connected in reverse polarity

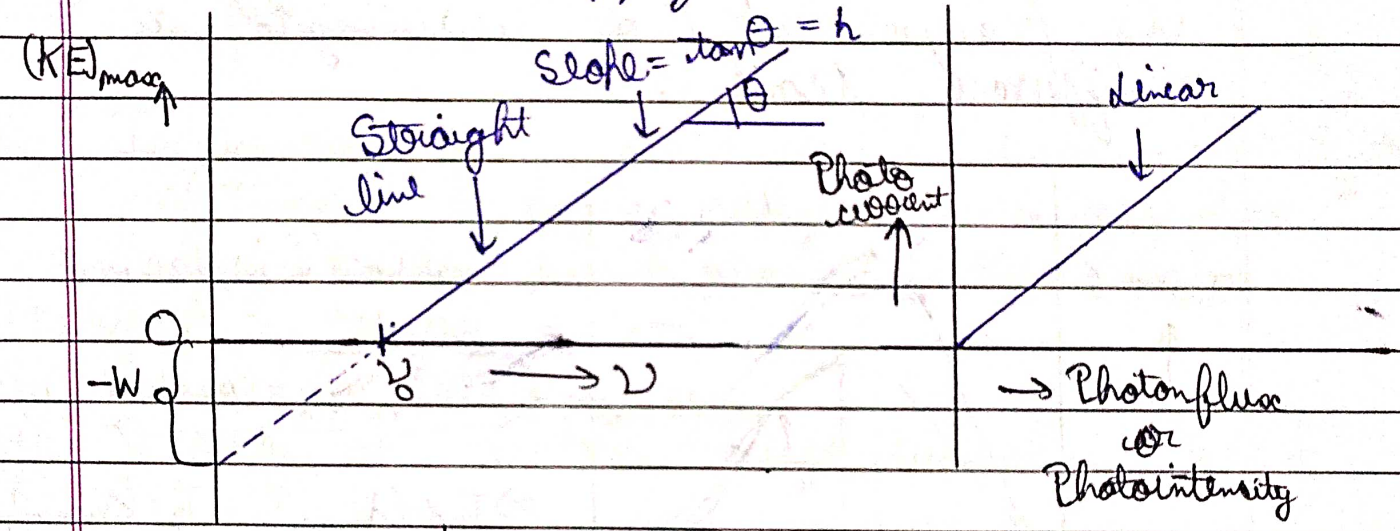


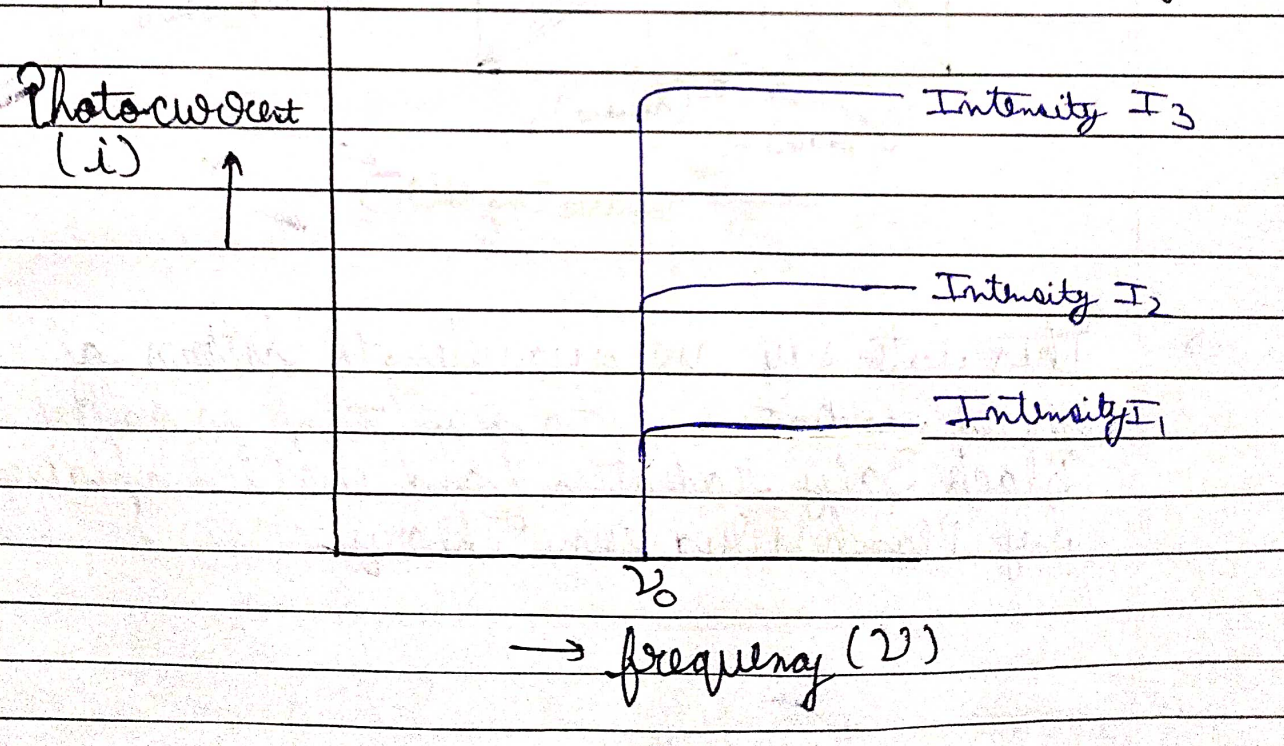
→ Stopping Potential (or Voltage) is the deaccelerating P.D. to be applied so that photo e^- with v_{max} KE are also stopped to reach to reciever plate i.e stopped the phenomenon of Photoelectric ~~current~~ effect.

$$q \times V = (KE)_{max}$$

$$\boxed{e \times V_0 = KE_{max} = h\nu - W}$$

where V_0 = Stopping Potential



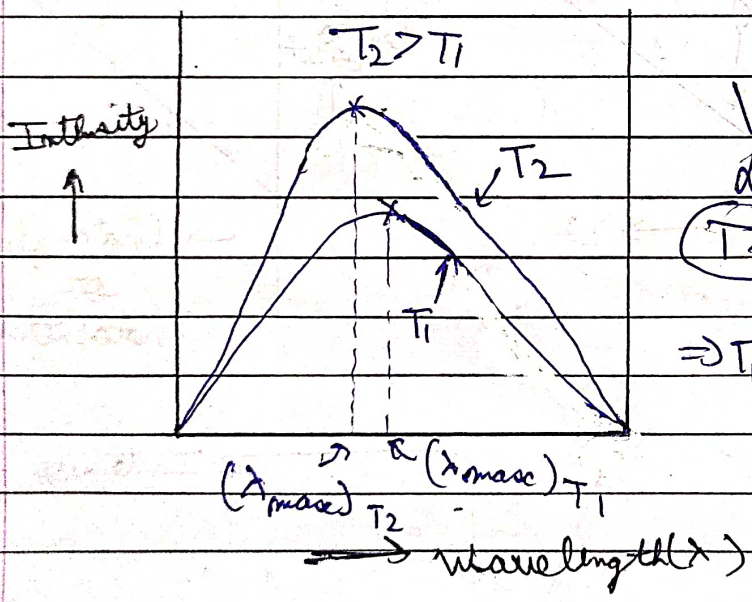


Black Body Radiation :-

Black body :- An ideal body which absorb and emit all frequencies is called Black body.

→ When an object is heat, emit radiations of wide variety of frequencies.

→ Here we study the pattern of intensity vs frequencies (or wavelengths) at different temp.



Wein's Displacement law

$$T \propto \lambda_{max} = \text{Constant}$$

$$\Rightarrow T_1 \times (\lambda_{max})_{T_1} = T_2 \times (\lambda_{max})_{T_2}$$

→ This intensity vs wavelength pattern of emitted radiations at a given temp is called Black Body radiation and well explained by Planck's Quantum Theory.

Q A monochromatic light of wavelength of 400 nm strikes a metal of work function 2 eV. Find
(i) Max KE of photoelectron
(ii) Threshold wavelength [1 nm = 10 Å]

Ans (i) $(KE)_{max} = \frac{hc}{\lambda} - W$

$= \frac{12400}{4000} \rightarrow = 3.1 - 2$

$= 1.1 eV$

(ii) $W = \frac{hc}{\lambda_0} \Rightarrow \lambda_0 = \frac{12400}{2} = 6200 Å$ Ans

Q In an experiment on a metal, when wavelength of Incident light is changed from 2000 Å to 1600 Å. KE of photoelectron gets doubled, find
(i) Work function of metal
(ii) Threshold frequency (Photo)
(iii) Velocity of electron with wavelength 2000 Å.

Ans $\frac{KE_1}{2KE_1} = \frac{\frac{hc}{2000} - W}{\frac{hc}{1600} - W} \Rightarrow \frac{1}{2} = \frac{\frac{12400}{2000} - \frac{12400}{\lambda_0}}{\frac{12400}{1600} - \frac{12400}{\lambda_0}}$

$\frac{1}{1600} = \frac{1}{\lambda_0} = \frac{1}{1000} - \frac{2}{\lambda_0}$

$\frac{1}{\lambda_0} = \frac{1}{1000} - \frac{1}{1600}$

$\Rightarrow \lambda_0 = \frac{8000}{3} Å$

(i) $W = \frac{12400}{(8000/3)} = \frac{12.4 \times 3}{8} = \frac{9.3}{2} = 4.65 eV$ Ans

(ii) $\nu_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{\frac{8}{3} \times 10^{-7}} Hz = \frac{9}{8} \times 10^{15} Hz$ Ans

(iii) $(KE)_{max} = \frac{12400}{2000} - 4.65$

$= 6.2 - 4.65$

$= 1.55 eV$

$\frac{1}{2} \times 9.1 \times 10^{-31} \times v^2 = 1.55 \times 1.6 \times 10^{-19}$

$\Rightarrow v^2 = \frac{1.55 \times 1.6 \times 2 \times 10^{12}}{9.1}$

$\Rightarrow v^2 = 7.4 \times 10^5 \text{ m/s}$

Q A monochromatic light of wavelength 310nm is irradiated on a metal of work function 2.5eV.
Find (i) Stopping Potential
(ii) Threshold Wavelength.

Ans (i) $eV_0 = (KE)_{max} = \frac{hc}{\lambda} - W$

$eV_0 = \frac{12400}{3100} - 2.5$

$= 4 - 2.5 = 1.5 eV$

$\Rightarrow$ Stopping Potential $V_0 = 1.5$ Volt Ans

(ii) $W = \frac{hc}{\lambda_0} \Rightarrow \lambda_0 = \frac{12400}{2.5} = 12400 \times \frac{4}{10}$

$= 4960 \text{ Å}$ Ans
(Visible light)

Imp

Q A light source rated as 16 watt emits light of wavelength 310 nm when this light falls completely on a metal of work function 2·5eV, then find photocurrent if only 50% of Photocurrent are ejected

Ans $\underbrace{16\times 1}_{\text{Joule}} = x \times \frac{12400}{3100} \times \underbrace{1.6\times 10^{-19}}$

$\Rightarrow x = \frac{1}{0.4}\times 10^{19}$

$\Rightarrow x = 2.5\times 10^{19}$ Photon / Sec

No of Photoelectrons ejected from metal

$= \frac{1}{2}\times 2.5\times 10^{19}$

$= 1.25\times 10^{19}$ Photo e^- / Sec

Current $= 1.25\times 10^{19}\times 1.6\times 10^{-19}$ Coulomb / Sec

$= 2$ amp. Ans

Bohr's Atomic Model :-

(Based on Planck's Quantum Theory and Newtonian's Physics

Postulates : (1) Atom consists of a dense tvely charged nucleas and e^- revolve around it in circular orbits. Force of Attraction is balanced by centrifuge force

(2) Out of infinite possible orbits, e^- revolves only in those orbits where its angular momentum is integral multiple of $\frac{h}{2\pi}$.

$\Rightarrow \underbrace{mvr}_{\text{Angular Momentum of } e^-} = n \times \frac{h}{2\pi}$ where $n = 1, 2, 3, 4 \cdots$ ← Orbit number

$n = 1, 2, 3, 4, \cdots$

Orbit → K, L, M, N, ---

Designation

These orbits where angular momentum is quantised are called stationary orbits or Energy levels where electron doesn't lose its energy.

→ Angular momentum of e^- in these orbits may be $1 \times \frac{h}{2\pi}$, $2 \times \frac{h}{2\pi}$, $10 \times \frac{h}{2\pi}$ but not the fractional multiple of $\frac{h}{2\pi}$.

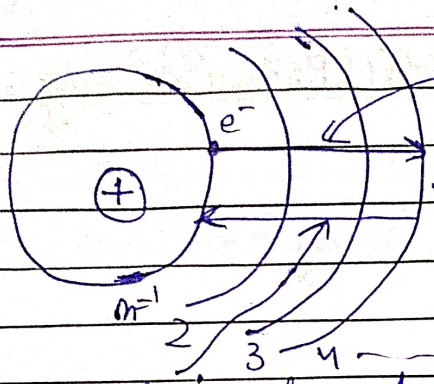
(3) Each orbit is associated with a definite amount of Energy. Energy of orbit increases as the orbit no. rises.

$E_1 < E_2 < E_3 < \cdots$

Imp (4) Energy is absorbed or emitted only when electron jumps ~~for~~ from one stationary orbit to another.

→ After absorbing energy (light or heat) electron will jump from lower to higher orbit.

→ Energy is emitted in the form of light or photon when electron jumps down to lower Energy level.



after absorbing energy

e⁻ jumps down to gain stability and emits light.

Energy of light absorbed or emitted = The energy difference of energy levels in transition

→ If e⁻ jumps down, then energy of photon emitted,

$$\frac{hc}{\lambda} = E_{n_2} - E_{n_1}$$

$n_2 \longrightarrow n_1$ (transition)

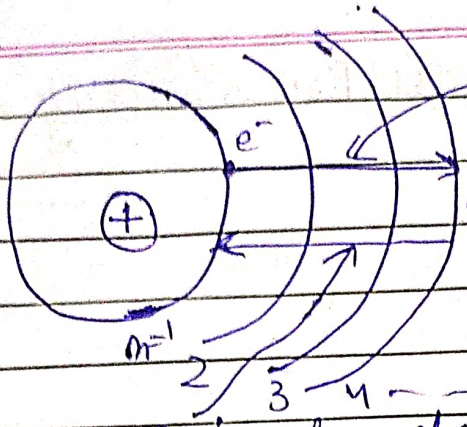
* One transition is equivalent to one photon.

Note :- Bohr's Atomic Model satisfactorily explains spectrum of single e⁻ species. like H, He^+, Li^{2+}, Be^{3+}, ---, etc.

⇒ Bohr model is applicable for single e⁻ species and fails to explain spectra of multi e⁻ species.

Q. Bohr's model is not applicable for

(A) H (B) He (3) C^{5+} (D) Li^{2+}



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Expression Of Orbit Radius & Velocity of e^- :-

$$\Rightarrow \boxed{mvr = \frac{nh}{2\pi}} \quad -(1)$$

$$\Rightarrow \frac{Kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$\Rightarrow \frac{mv^2}{r} = \frac{K(e)(+Ze)}{r^2}$$

$$\Rightarrow \boxed{\frac{mv^2}{r} = \frac{KZe^2}{r^2}} \quad -(2)$$

$$V = \frac{nh}{2\pi mr}$$

$$\frac{m}{r} \times \frac{n^2h^2}{4\pi^2m^2r^2} = \frac{KZe^2}{r^2}$$

$$\Rightarrow r = \frac{n^2h^2}{4\pi^2mKZe^2} = \frac{(6.625\times10^{-34})^2 \times n^2}{4\pi^2 \times 9.1\times10^{-31} \times 9\times10^9 \times (1.6\times10^{-19})^2}$$

$$\Rightarrow r_n = 0.529 \times \frac{n^2}{Z} \times 10^{-10}\ m$$

$$\Rightarrow \boxed{r_n = 0.529 \times \frac{n^2}{Z}\ \text{Å}}$$

$$mvr = \frac{nh}{2\pi}$$

$$V = \frac{nh}{2\pi} \times \frac{1}{mr}$$

$$= \frac{nh}{2\pi} \times \frac{1}{mn^2h^2/4\pi^2mKZe^2}$$

$$= \frac{2\pi mKZe^2}{nh}$$

$$\Rightarrow \boxed{V_n = 2.188 \times 10^6 \frac{Z}{n} \text{ m/sec}}$$

x x

$$\Rightarrow \boxed{r_n = 0.529 \times \frac{n^2}{Z} \text{ Å}}$$

$$\boxed{V_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}}$$

x x

Expression of Time Period & Frequency of Revolution:

Time Period of e⁻: Time taken by the e⁻ to complete one revolution

$$V_n = \frac{2\pi r_n}{T_n}$$

$$\Rightarrow \boxed{T_n = \frac{2\pi r_n}{V_n}}$$

$$T_n \propto \frac{n^2/Z}{Z/n}$$

$$\Rightarrow \boxed{T_n \propto \frac{n^3}{Z^2}}$$

Frequency of revolution of e⁻ (f_n): No of revolution per sec

$\because$ T sec ——— = 1 rev

$\therefore$ 1 sec ——— $= \frac{1}{T_n} = f_n$.

$$\Rightarrow f_n = \frac{1}{T_n} = \frac{V_n}{2\pi r_n}$$

$$f_n \propto \frac{Z/n}{n^2/Z}$$

$$\boxed{f_n \propto \frac{Z^2}{n^3}}$$

Q Find the ratio of radii of circular orbits of 4th orbit of He^+ & 2nd orbit of H.

Ans $r_n = 0.529 \times \frac{n^2}{Z}$

$r_n \propto \frac{n^2}{Z}$

$\frac{(r_4)_{He^+}}{(r_2)_H} = \frac{(4^2/2)}{(2^2/1)}$

$= \frac{8}{4} = 2:1$ Ans

Q Find the ratio of velocity of e^- in 2nd orbit of Li^{2+} and 3rd orbit of He^+.

Ans $V_n \propto \frac{Z}{n}$

$\frac{(V_2)_{Li^{2+}}}{(V_3)_{He^+}} = \frac{(3/2)}{(2/3)} = \frac{9}{4}$ Ans

Q Find velocity of e^- in ground state ($n=1$) of He^+.

Ans $V_1 = 2.18 \times 10^6 \times \frac{Z}{n}$ m/sec

$= 2.18 \times 10^6 \times \frac{2}{1}$ m/sec

$= 4.36 \times 10^6$ m/sec Ans

Q Find the ratio of Time Period of e^- in 2nd orbit of H atom and 3rd orbit of He^+.

Ans $T_n = \frac{2\pi r_n}{V_n}$

$\propto \frac{n^2/Z}{Z/n}$

$\propto \frac{n^3}{Z^2}$

$\frac{(T_2)_H}{(T_3)_{He^+}} = \frac{2^3/1^2}{3^3/2^2}$

$= \frac{8 \times 4}{1 \quad 27}$

$= 32:27$ Ans

Data Page

Expression Of Energy :

$$TE = KE + PE$$

$$= \frac{1}{2} mv^2 + \frac{kq_1q_2}{r}$$

$$= \frac{1}{2} mv^2 - \frac{kZe^2}{r}$$

$$\therefore \frac{mv^2}{r} = \frac{kZe^2}{r^2}$$

$$\Rightarrow TE = \frac{kZe^2}{2r} - \frac{kZe^2}{r}$$

$$\Rightarrow \boxed{TE = -\frac{kZe^2}{2r}}$$

$$\Rightarrow \boxed{PE = 2 \times T.E = -2 \times KE}$$ ✓

$$T.E = E_n = -\frac{9 \times 10^9 \times Z \times (1.6 \times 10^{-19})^2}{2 \times 0.529 \times \frac{n^2}{Z} \times 10^{-10}}$$

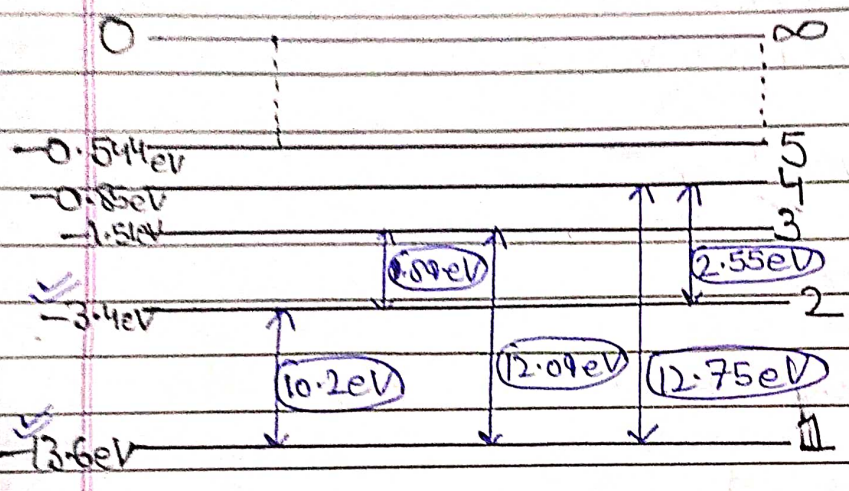
$$\Rightarrow \boxed{E_n = -13.6 \times \frac{Z^2}{n^2} \text{ in eV}}$$

For H-atom, $E_1 = -13.6\,eV$
$E_2 = -3.4\,eV$
$E_3 = -1.51\,eV$
$E_4 = -0.85\,eV$
$E_5 = -0.544\,eV$
⋮
$E_\infty = 0$ (No more attⁿ b/w e^- and nucleus)

discrete values indicate that Energy is quantised

V. Imp

Energy level diagram for atomic Hydrogen



⇒ Energy of photon emitted or absorbed

$$= \frac{hc}{\lambda} = E_{m_2} - E_{m_1}, \quad m_2 > m_1$$

Q — In atomic hydrogen, e^- undergoes transition from 3rd orbit to ground state, find wavelength of Photon emitted.

Ans $3 \longrightarrow 1$

$$E_3 - E_1 = \frac{hc}{\lambda}$$

$$\frac{12400}{\lambda} = 12.09$$

$$\lambda = \frac{12400}{12.09} \text{ Å}$$

$$= 1025.6 \text{ Å}$$ Ans

Q Find energy of photon emitted and also its wavelength when e^- jumps from 2nd orbit to 1st orbit in He^+ ion.

Ans $E_{n_2} - E_{n_1} = \frac{hc}{\lambda}$

$E_2 - E_1 = \frac{hc}{\lambda}$

$-\frac{13.6 \times 2^2}{2^2} + \frac{13.6 \times 2^2}{1^2} = \frac{12400}{\lambda}$

$13.6 \times 3 = \frac{12400}{\lambda}$

$\lambda = \frac{12400}{13.6 \times 3}$ A^0 Ans

Alternatively,

$E_2 - E_1 = \frac{hc}{\lambda}$ | $E_n = \frac{-13.6 \times Z^2}{n^2}$

$10.2 \times Z^2 = \frac{hc}{\lambda}$ | $E_n = (E_n)_H \times Z^2$

$\lambda = \frac{12400}{10.2 \times 2^2}$

$= \frac{12400}{40.8}$ Å Ans

Some Definitions for single e^- species:

1. Ground State: $n=1$ (most stable state)
2. Excited State: Any orbit other than $n=1$

1st excited state: $n=2$
2nd excited state: $n=3$
⋮
x^{th} excited state: $n=(x+1)$

(3) **Ionisation Energy (IE)** :- Minimum energy needed to remove e^- from ground state to ∞.

$$IE = E_\infty - E_1 = -(-13.6 \times Z^2)$$

$$\boxed{IE = 13.6 Z^2 \text{ eV}}$$

$\Rightarrow (IE)_H = 13.6 \text{ eV}$

$\Rightarrow (IE)_{He^+} = 54.4 \text{ eV}$

(4) **Excitation Energy** : Energy required to excite electron from ground state to a required excited state.

1^{st} excitation energy $= E_2 - E_1$

2^{nd} excitation energy $= E_3 - E_1$

$\vdots$

6^{th} excitation energy $= E_7 - E_1$

(5) **Ionisation Potential (IP)** : It is potential difference to be applied on e^- at rest to get its KE equal to Ionisation Energy.

$$q \times V = IE = 13.6 Z^2 \text{ eV}$$

$$1e \times V_{IP} = 13.6 Z^2 \text{ eV}$$

$$\boxed{V_{IP} = 13.6 Z^2 \text{ volt}}$$

(6) **Binding Energy** :- It is the energy required to remove electron from any excited state to ∞.

Eg: B.E of 1st excited state $= E_\infty - E_2$

$= -\left(-13.6 \times \frac{Z^2}{2^2}\right)$

$= 3.4 Z^2$ Ans

Q 1st Excitation energy of e^- in H atom is x eV. Find 2nd excitation energy of e^- in He^+ ion in terms of x.

Ans

$E_2 - E_1 = x$

$13.6 \times 1^2 \left(\frac{1}{1^2} - \frac{1}{2^2}\right) = x$

$13.6 \times \frac{3}{4} = x$

2nd excitation energy in He^+

$= E_3 - E_1$

$= 13.6 \times 2^2 \left(\frac{1}{1^2} - \frac{1}{3^2}\right)$

$= \frac{4x}{3} \times 4 \left(\frac{8}{9}\right)$

$= \frac{128x}{27}$ Ans

Q 2nd excitation energy of a hypothetic species following Bohr model is x eV. Find

(i) I.E

(ii) Binding Energy of 2nd orbit

Ans

$E_n = \frac{-K \times 1}{n^2}$ [$13.6 Z^2 = K$]

$E_3 - E_1 = x \Rightarrow K\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = x$

$K = \frac{9x}{8}$

(i) $IE = K = \frac{9x}{8}$ eV

(ii) Binding energy of 2nd orbit $= E_\infty - E_2$

$= 0 + \frac{K}{2^2} = \frac{9x}{8 \times 4} = \frac{9x}{32}$ Ans

Q Binding of H like system corresponding to 2nd excited state is 13.6 eV
(i) Identify the sample.
(ii) Find IE
(iii) Find wavelength of photon when e jumps down from $n=2$ to $n=1$.

Ans $E_\infty - E_3 = 13.6 \Rightarrow 13.6 \times \frac{Z^2}{3^2} = 13.6$

(i) $Z = 3 \ (Li^{2+})$ Ans

(ii) $IE = 13.6\, Z^2 = 13.6 \times 9 = 122.4\ eV$

(iii) $E_2 - E_1 = \frac{hc}{\lambda} \Rightarrow \frac{12400}{\lambda} = 10.2 \times 3^2$

$\lambda = \frac{12400}{10.2 \times 9}\ \text{Å}$

$= \frac{12400}{91.8}\ \text{Å}$ Ans

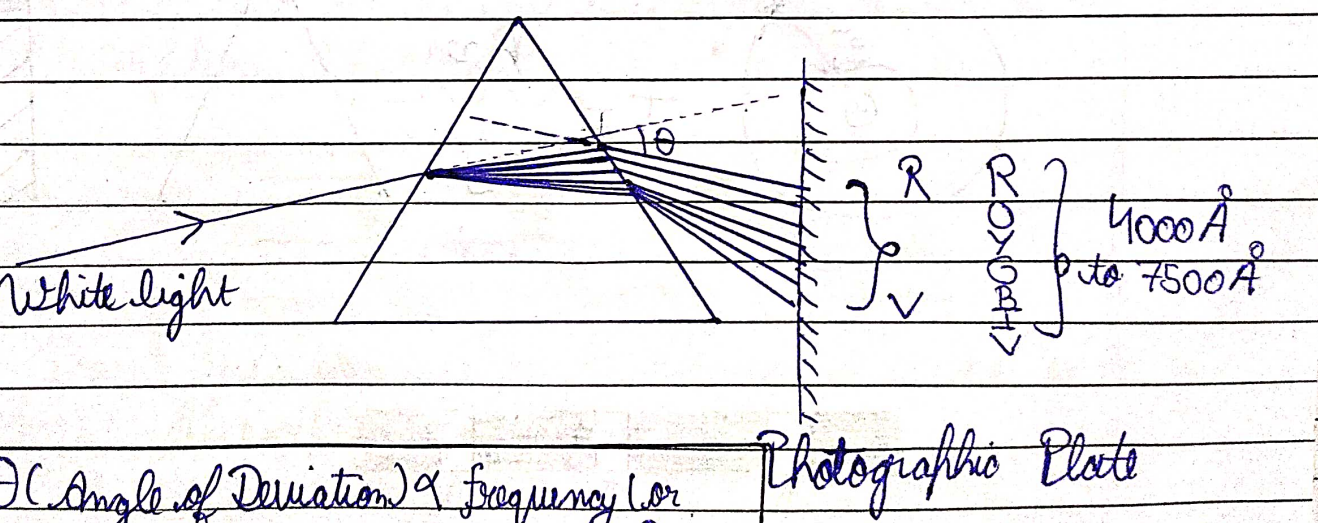
Spectrum:- When a light from a source is passed through a prism light of different wavelengths get deviated through different angles. This is called dispersion. This wavelength composition of dispersed light can be viewed by eye when received on a photographic plate. This wavelength composition is called Spectrum.

Types of Spectrum

- Emission Spectrum
  - Continuous
  - Discontinuous
- Absorption Spectrum

(A) **Emission Spectrum:** When a light emitted by a source is dispersed and recieved on Photographic plate, the formed wavelength pattern is called Emission Spectrum.

(1) **Continious Spectrum :-** When white light (eg Sunlight or light from tubelight/bulbs) is dispersed through a prism and recieved on dark background (photo graphic plate), the entire photographic plate where light falls become bright consisting seven colours (VIBGYOR). The intensity of various colours continiously change and there is no sharp boundary b/w these colours. Infact colours appear to be merging with next colour. Such spectrum of white light contain all wavelengths in a certain wavelength range (4000 Å to 7500 Å for white light)

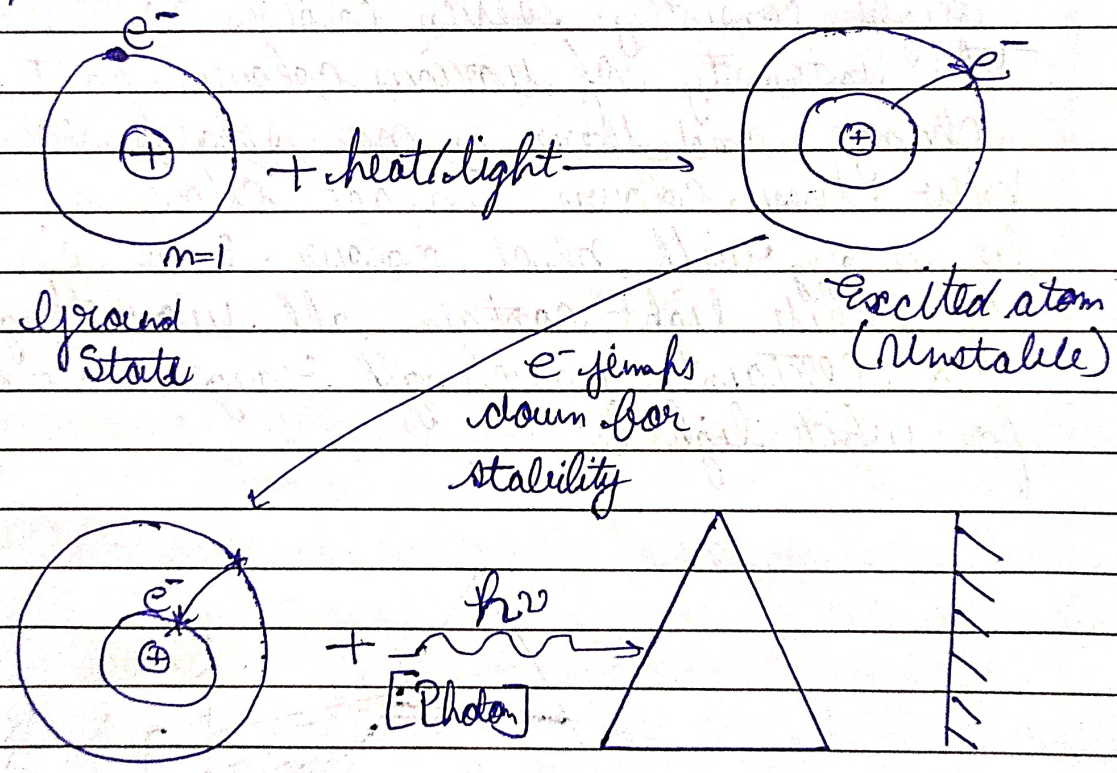


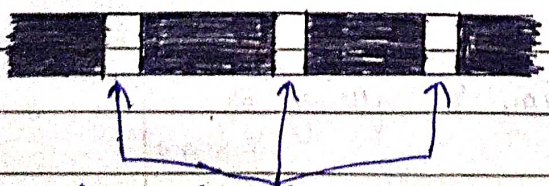
θ (Angle of Deviation) ∝ Frequency (or Energy)

$$\propto \frac{1}{\lambda}$$

(ii) Discontinious Spectrum:

(a) Line Spectrum: When atoms of an element are heated (or irradiated by light), atoms get excited and e^- jumps higher in various energy levels and become unstable and to gain stability it jumps down and emit light. This emitted light is dispersed and the pattern of dispersed light is line spectrum.

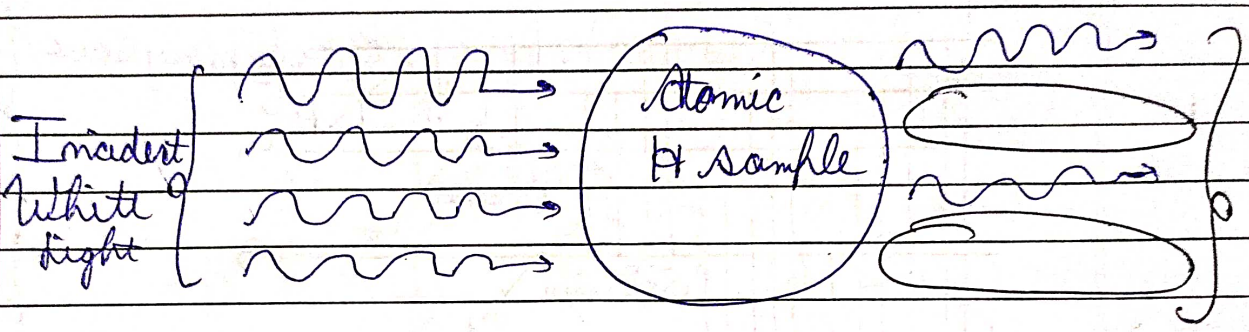


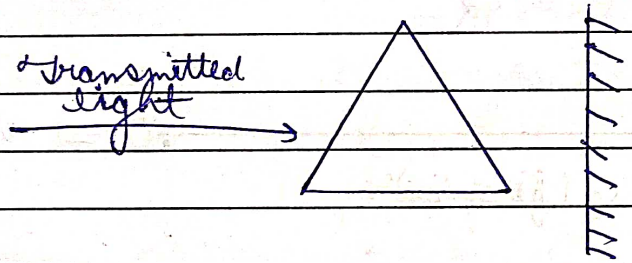


These bright lines on dark background is called line spectrum.

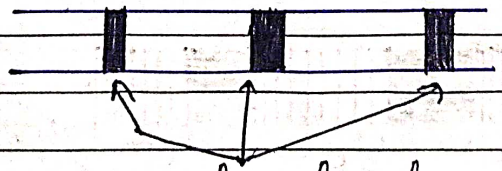
(b) Band Spectrum :- Observed in complex molecules
→ Band Spectrum is ~~actually~~ in fact line spectrum (discontinious) where lines are very closely spaced.

(B) Absorption Spectrum :-





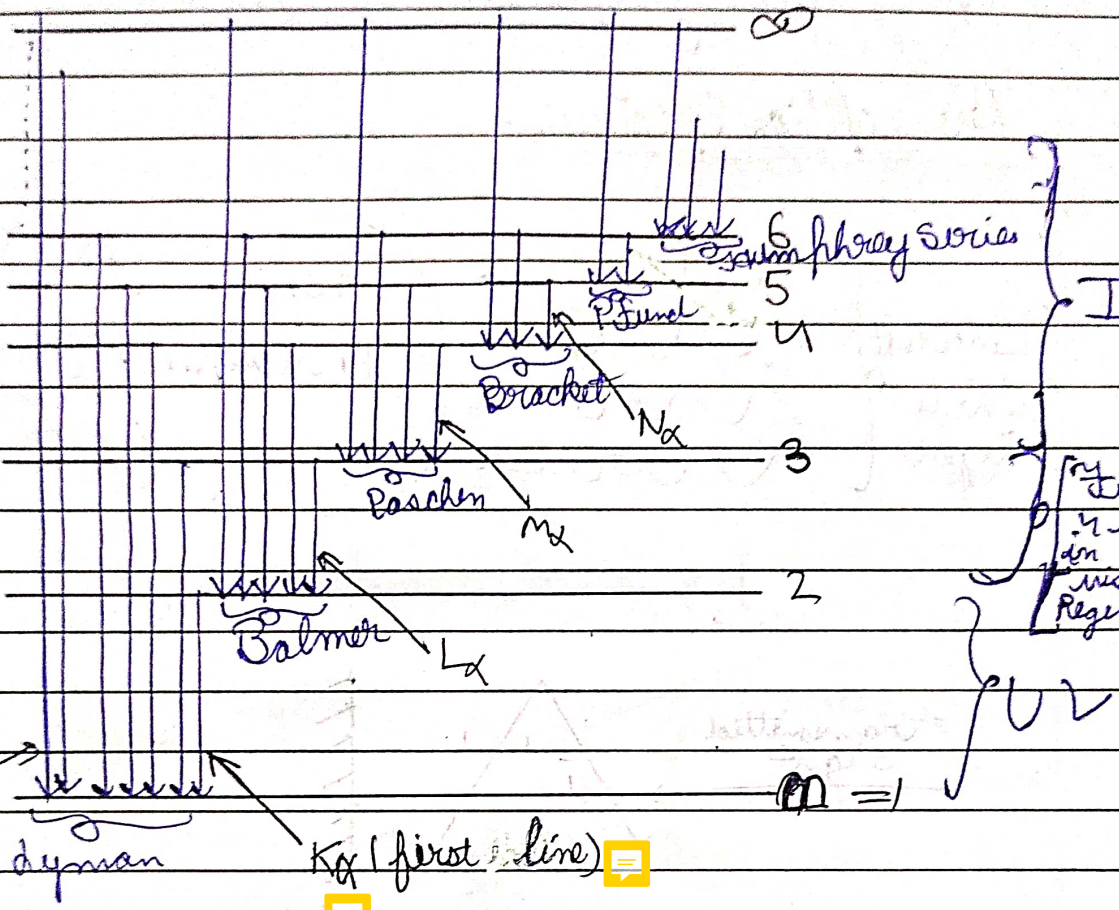
→ Dark lines on bright background is called Absorption Spectrum.



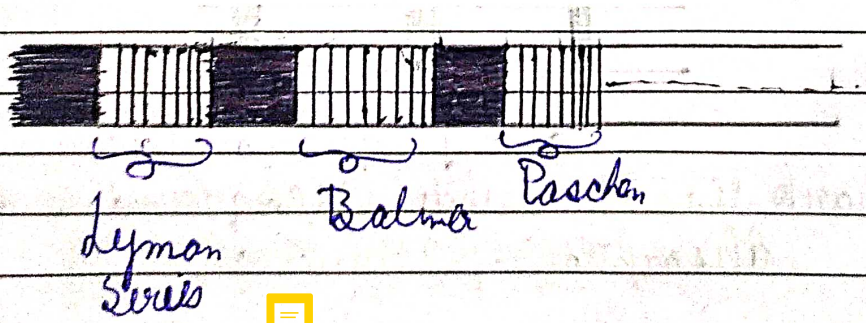
Dark lines on white background is called Absorption Spectrum.

Note: These dark lines on bright in absorption spectrum are same as bright lines in emission spectrum.

Line Spectrum Of Atomic Hydrogen :-



* Last line is also called series limit of any series ..



Emb No two elements have identical spectrums

→ If transition occurs from a higher energy level to a lower energy level then

$$E_{n_2} - E_{n_1} = \frac{hc}{\lambda}$$

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{hc}\left[\frac{-13.6z^2}{n_2^2} + \frac{13.6\,z^2}{n_1^2}\right]$$

$$\Rightarrow \bar{\nu} = \frac{1}{\lambda} = \left[\frac{13.6 \times z^2}{hc}\right]\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\Rightarrow \bar{\nu} = \frac{1}{\lambda} = R_H z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

where R_H = Rydberg's constant
$= 1.0967 \times 10^7$ meter^{-1}

For atomic Hydrogen, $z = 1$

$$\frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

(i) For lyman series: $n_1 = 1$, $n_2 = 2, 3, 4, \ldots \infty$

$$\frac{1}{\lambda} = R_H\left(1 - \frac{1}{n_2^2}\right)$$

For 1st line (K_α) of lyman series, $n_2 = 2$

$$\frac{1}{\lambda_{max}} = R_H\left(1 - \frac{1}{4}\right) = \frac{3R_H}{4}$$

$$\lambda_{max} = \frac{4}{3R_H} = \underline{1216\,\text{Å}}$$

For last line of lyman series, $n_2 = \infty$ [series limit]

$$\frac{1}{\lambda_{min}} = R_H\left(1 - \frac{1}{\infty^2}\right) \Rightarrow \lambda_{min} = \frac{1}{R_H}$$

$$= 912\,\text{Å}$$

→ All wavelengths in lyman series are in UV range.

(ii) Balmer Series: $n_1 = 2,\ n_2 = 3, 4, 5, \cdots, \infty$

$$\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{n_2^2}\right)$$

For first line of Balmer series, $n_2 = 3$

$$\frac{1}{\lambda_{max}} = R_H\left(\frac{1}{4} - \frac{1}{9}\right)$$

$$\lambda_{max} = \frac{36}{5R_H} = 6563\text{Å}\ (\text{visible region})$$

For last line of Balmer Series, $n_2 = \infty$

$$\frac{1}{\lambda_{min}} = R_H\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$$

$$\lambda_{min} = \frac{4}{R_H}$$

$$= 3646\text{Å}\ (\text{UV Range})$$

Imp.

→ First four line of Balmer series ($3 \to 2$, $4 \to 2$, $5 \to 2$, $6 \to 2$) fall in visible region range (4000Å to 7500 Å) and remaining all lines fall in UV range.

(iii) Paschen Series: $n_1 = 3, n_2 = 4, 5, 6, \ldots \infty$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{3^2} - \frac{1}{n_2^2}\right)$$

(iv) Brackit Series
(v) Pfund Series
(vi) Humphrey Series

→ fall in IR region.

Calculation of no of Spectral lines:

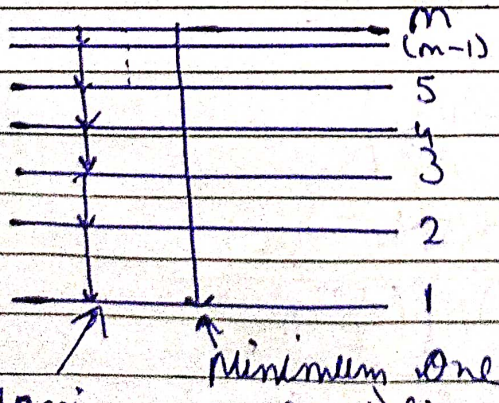
⇒ Max. No of Spectral lines obtained when electron undergo transition from n^{th} orbit to ground state in a sample of H-like Spectrum

$$= \frac{n(n-1)}{2} \text{ or } {}^nC_2$$

⇒ No of Spectral lines when transition occurs from n_2 to n_1 in a sample of H-like species.

$$= \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

⇒ Max No. of Spectral lines when an isolated H-like species undergo transition from n^{th} orbit to ground state $= \frac{(n-1)}{}$



* Minimum no. of Spectral lines from n^{th} orbit to $n=1$
$= 1$

Q Find wavelength of 2nd line of Balmer series in terms of Rydberg's ~~Energy~~ constant for He^+

Ans $4 \to 2$

$$\frac{1}{\lambda} = R_H \times 2^2 \left[\frac{1}{2^2} - \frac{1}{4^2}\right]$$

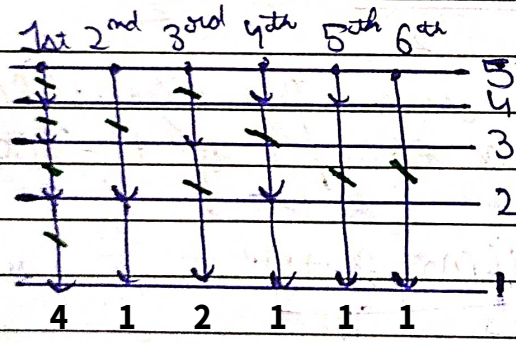
$$\frac{1}{\lambda} = 4 \times R_H \left[\frac{4-1}{16}\right] = \frac{3R_H}{4}$$

$$\boxed{\lambda = \frac{4}{3R_H}}$$ Ans

IMP

Q Find minimum no of H atoms needed to show 10 Spectral lines if all atoms are present in 4th excited state initially.

Ans



Minimum 6 atoms of H are needed to show 10 lines from 5th orbit to ground state.

Q Find ratio of wave number of 2nd line of Lyman series of He^+ ion and series limit of Balmer series of H atom.

Ans

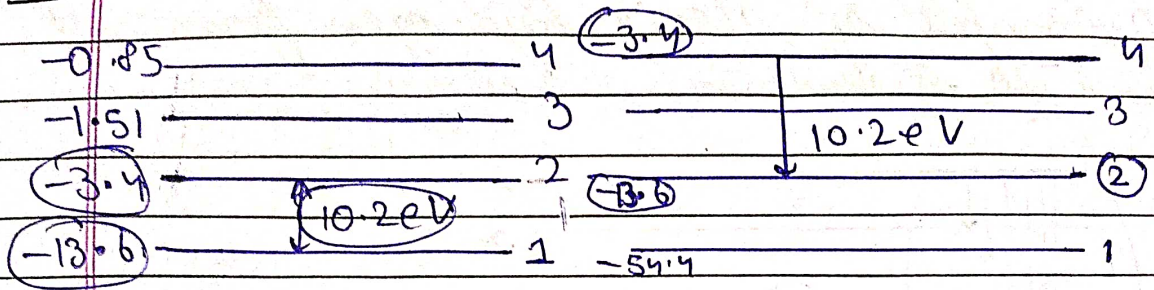
$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{(\bar{\nu}_{He})_{3\to1}}{(\bar{\nu}_{H})_{\infty\to2}} = \frac{R_H \times 4 \left(\frac{1}{1^2} - \frac{1}{3^2}\right)}{R_H \times 1 \left(\frac{1}{2^2} - \frac{1}{\infty}\right)}$$

$$= \frac{4 \times \frac{8}{9}}{1 \times \frac{1}{4}} = \frac{128}{9}$$ Ans

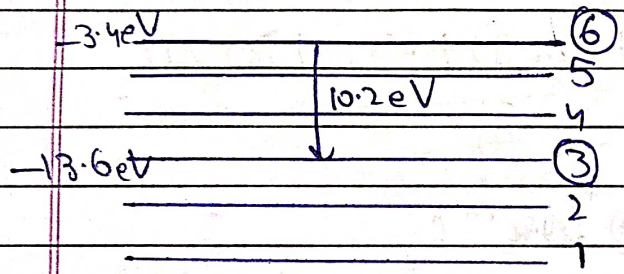
V. Imp

Note :



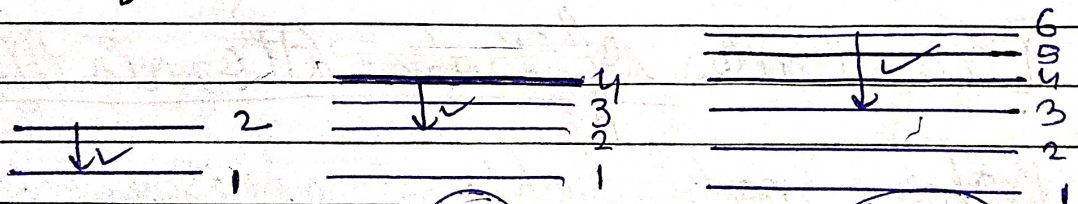
H
Z=1

He+
Z=2



Li^{2+}
Z=3

Q ★ A sample of all hydrogens in 1st excited state, add He^+ in 3rd excited state, and all Li^+ in 5th excited state. Find max no of transitions that can be obtained.



H He^+ Li^{2+}

$\frac{1}{=}$ $\frac{4\times3}{2} = 6 - 1$ $\frac{6\times5}{2} = 15 - 1$

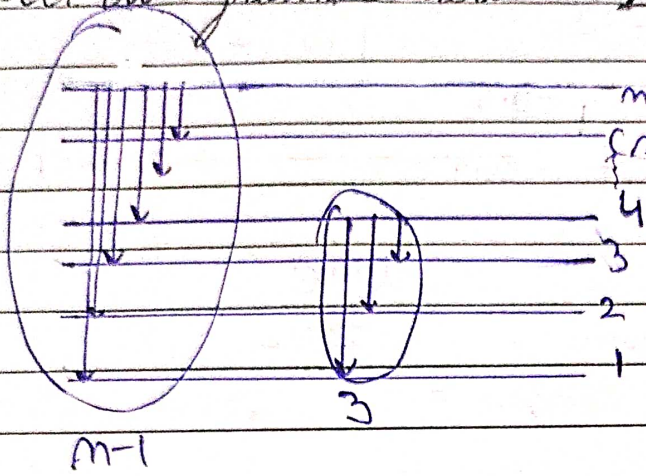
$= 5$ $= 14$

Max. Spectral lines = 1 + 5 + 14

= 20 Ans

Q Prove that max no of spectral lines in a sample of atomic hydrogen from n^{th} orbit to ground state = $\frac{n(n-1)}{2}$.

Ans



$$= \underbrace{(n-1)+(n-2)+(n-3)+\cdots+3+2+1}_{n \text{ terms}}$$

$$= \frac{n(n-1)}{2} \text{ or } {}^{n}C_{2}$$

Q Find max no of spectral lines in visible region ~~in~~ range in H sample if all atoms are in 6th excited state.

Ans $7 \longrightarrow 1$

$\underline{4}$ Ans $[3\to2, 4\to2, 5\to2, 6\to2]$

Drawbacks Of Bohr Atomic Model :-

(1) Bohr model explains spectrum of single e^- species only (like H, He^+, Li^{2+} etc)

It fails to explain spectra of multi e^- species (like He, Na, Fe etc)

* It is due to shielding effect or screening effect in multi e^- species

(2) When a high resolution power spectroscope is used, then each spectral line in atomic hydrogen split into two fine lines (Doublet). Bohr couldn't explain the doublet formed or fine spectrum of atomic hydrogen.

* later this was attributed to the spin of an electron about its own axis.

(3) Bohr could not explain the splitting of spectral lines in influence of magnetic field (called Zeeman effect) and in Electric field (called Stark effect).

(4) According to Bohr e revolves in circular orbits and around nucleas. But Heisenberg concluded that path or trajectory of a moving microscopic particle like e⁻ can not be determined.

De-Broglie's Hypothesis:

De-Broglie proposed that particle as well as wave nature of moving microscopic particles like electron, proton, atom etc

Davison and Germen determined the wave nature of fast moving electrons and observed the spectrum (or wave nature) of moving e⁻ similiar to X-rays. Also the wavelength of wave nature of e⁻ was found to

matching with wavelength expression given by de-Broglie.

This validates the wave nature of moving e^-.

Energy $\Rightarrow E = mc^2 = \frac{hc}{\lambda}$ [light or Photon]

$$\lambda = \frac{h}{mc} \quad \text{(for Photon)}$$

$$\lambda = \frac{h}{p} \quad \text{(for Photon)}$$

By analogy, de Broglie proposed,

$$\boxed{\lambda = \frac{h}{mv}}$$ → for moving matter particles

De Broglie wavelength

m = mass of matter particles

v = velocity

$$\Rightarrow \boxed{\lambda = \frac{h}{mv}} \quad \text{(1)}$$

$$mx\ KE = \frac{1}{2} m^2v^2 \Rightarrow mv = \sqrt{2mKE}$$

$$\Rightarrow \boxed{\lambda = \frac{h}{\sqrt{2m(KE)}}} \quad \text{(2)}$$

KE of a charged particle under P.D. of V volt,

$\Rightarrow KE = q \times V$

$$\boxed{\lambda = \frac{h}{\sqrt{2m \times q \times V}}}$$

$$\Rightarrow \lambda_e = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times V}} \text{ meter}$$

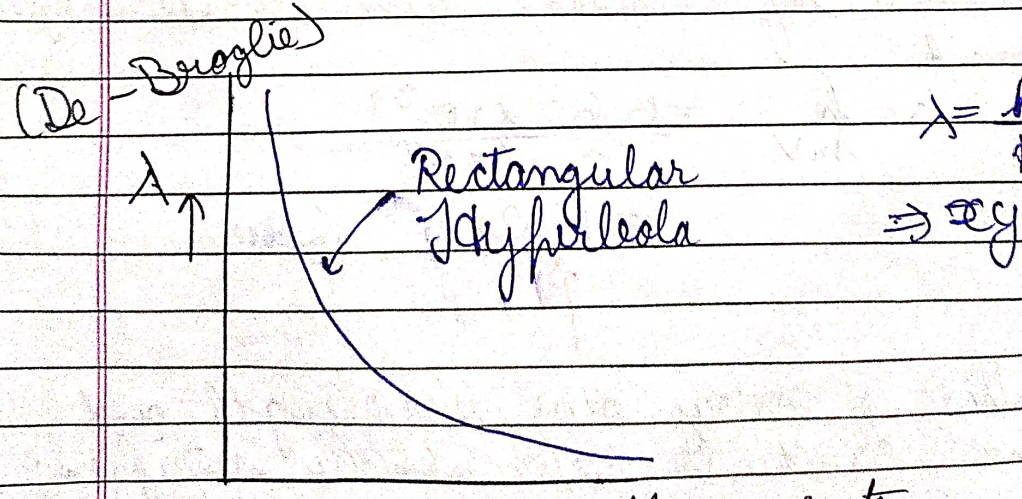
$$\lambda_e = \sqrt{\frac{150}{V}} \times 10^{-10} \text{ meter}$$

$$\Rightarrow \boxed{\lambda_e = \sqrt{\frac{150}{V}} \text{ Å}}$$

Summary :- $\lambda = \frac{h}{mv} = \frac{h}{p}$

$$\lambda = \frac{h}{\sqrt{2m(KE)}}$$

$$\lambda = \frac{h}{\sqrt{2m \times q \times V}}$$



$\lambda = \frac{h}{p}$

$\Rightarrow xy = \text{constant}$

Q Find ratio of de-Broglie wavelength of photon and α-particle
(i) If both have same KE
(ii) If both are accelerated from rest by same P.d.

Ans (i) $\lambda = \frac{h}{\sqrt{2mKE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{m}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha}{m_p}}$$

$$= \sqrt{\frac{4 \times m_p}{m_p}}$$

$$= 2 \text{ Ans}$$

(ii) $\lambda = \frac{h}{\sqrt{2mqV}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha \times q_\alpha}{m_p \times q_p}}$$

$$= \sqrt{\frac{4 \times 2e}{1 \times e}} = 2\sqrt{2} \text{ Ans}$$

Q A ball of mass 100 g is moving with a speed of 10 m/sec. Find its de-Broglie wavelength.

Ans $\lambda = \frac{h}{mV} = \frac{6.62 \times 10^{-34}}{0.1 \times 10}$

$= 6.62 \times 10^{-34}$ meter

This is very low wavelength and hence can't be experimentally verified.

⇒ de-Broglie's wave nature of macroscopic or large particle is insignificant.

JMA

Q A light of wavelength 310 nm is irradiated on a clean metal of work function 2eV. Find de-Broglie wavelength of most energetic photo-electron.

Ans $(KE)_{max} = \frac{hc}{\lambda} - W$

$= \frac{12400}{6200} - 2 = 2eV$

$V = 2$ volt, $\lambda_{e^-} = \sqrt{\frac{150}{V}} = \sqrt{\frac{150}{2}}$

$= \sqrt{75}\ \text{Å}$

Ans

Q Electron is revolving in 3rd orbit of Li^{2+} ion. Find its de-Broglie's wavelength.

Ans 1st Approach $\lambda = \frac{h}{mv}$

$(v_3)_{Li^{2+}} = 2.18 \times 10^6 \times \frac{Z}{n}$

$= 2.18 \times 10^6 \times \frac{3}{3}$

$= 2.18 \times 10^6$ m/s

$\lambda = \frac{6.62 \times 10^{-34}}{9.1 \times 10^{-31} \times 2.18 \times 10^6}$

$= \frac{6.62}{9.1 \times 2.18} \times 10^{-9}$ meter

$= \frac{66.2}{9.1 \times 2.18}$ Å

2nd Approach

Li^{2+}, $n=3$,

$$E_n = -13.6 \times \frac{Z^2}{n^2}$$

$$= -13.6 \times \frac{3^2}{3^2} = -13.6\,V$$

$$KE = |E_n| = 13.6\,eV$$

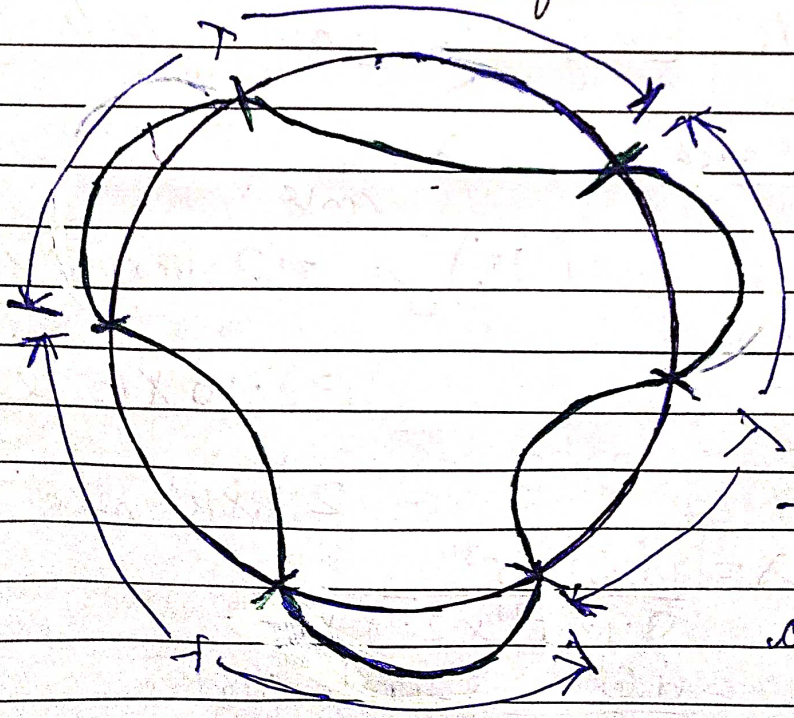
$$V = 13.6\ Volt$$

$$\lambda_e = \sqrt{\frac{150}{13.6}} = \sqrt{11} \simeq 3.3\,\text{Å}$$

Verification of Bohr's Angular Momentum Quantisation rule: (By de-Broglie)

$$mvr = n \times \frac{h}{2\pi}$$

→ Constructive Interference:



Constructive Interference (or in-phase combination of waves)

→ $2\pi r = n \times \lambda$

↑ Orbit number or no. of waves.

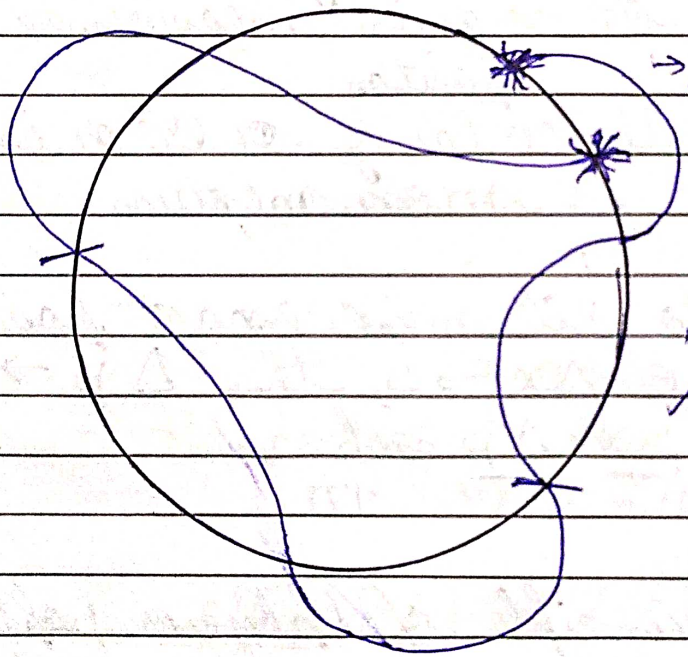
$2\pi r = n \times \lambda$

n ↑ no of waves or orbit number

$2\pi r = n \times \frac{h}{mv}$

$\boxed{mvr = n \times \frac{h}{2\pi}}$ → Bohr's Angular momentum quantisation rule.

→ Destructive Interference



→ Destructive Interference or out of phase combination

$\times\ 2\pi r \neq n\lambda$

Q Bohr's ground state radius of H atom is x Å. Find de-Broglie wavelength in 3rd orbit of He^+ ion in terms of x.

Ans $x = \frac{0.529 \times 1^2}{1}$ Å

$2\pi (r_3)_{He^+} = 3\lambda$

$\Rightarrow \lambda = \frac{2\pi}{3} \times 0.529 \times \frac{3^2}{1}$

$= 3\pi x$ Ans

Heisenberg's Uncertainity Principle :

→ According to Heisenberg, it is impossible to determine position and momentum of a moving microscopic particle (like e^-, p or atom) exactly (accurately) and simultaneously.

→ $$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

where Δx = Uncertainity in position or error in measurement of position

Δp = Uncertainity or error in momentum

→ If you try to determine exact position of e^-, i.e $\Delta x \to 0$, then $\Delta p \to \infty$ because $\Delta x \times \Delta p \geq \frac{h}{4\pi}$

→ A general principle of physics (optics) is that, the order of error in position of a particle is equal to wavelength of light employed to see the particle.

$\Delta x \cong \lambda$,

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$\Delta x \to 0$ (very accurate position)

$\Rightarrow \lambda \to 0$ | for light used

$$\boxed{\lambda = \frac{h}{mc}}$$

$$E = \frac{h\nu}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mc} = \frac{h}{p}$$

λ → Wavelength, p → Momentum of Photon

As a result momentum of electron will change by a large amount

$\therefore \Delta p \rightarrow$ very high

$$\rightarrow \Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \times m \Delta v \geq \frac{h}{4\pi} \quad \left[\begin{array}{l} p = mv \\ \Delta p = m\Delta v \end{array}\right]$$

$$\Rightarrow \underbrace{\Delta x \times \Delta v}_{\text{Uncertainity product where}} \geq \frac{h}{4\pi m}$$

Δv = Uncertainity or error in velocity.

$$\rightarrow \Delta x \times \boxed{\frac{\Delta p}{\Delta t}} \times \Delta t \geq \frac{h}{4\pi}$$

$$\rightarrow \boxed{F \times \Delta x} \times \Delta t \geq \frac{h}{4\pi}$$

$$\Delta E \times \Delta t \geq \frac{h}{4\pi}$$

ΔE → Uncertainity in measurement of Energy

Δt → Uncertainity or error in measurement of time

$\Delta x \times \Delta p \geq \frac{h}{4\pi}$ or $\Delta x \times \Delta p \geq \frac{\hbar}{2}$

where $\hbar = \frac{h}{2\pi}$

→ According to Heisenberg's Uncertainity principle, path or trajectory of moving microscopic particles like e^- can't be determined. This led to failure of Bohr's concept of circular orbits.

→ Heisenberg replaced the concept of orbit into the concept of orbital.

→ Heisenberg concluded that we can only find probability of finding e^- around nucleus at a point but can't find its exact location.

→ An orbital is the region of space around nucleus where probability of finding e^- are considerably high. (90 to 95%)

→ Heisenberg's Uncertainity principle is meaning less or insignificant for large particles.

$\Delta x \times \Delta v \geq \frac{h}{4\pi m}$ ↑

↳ small if m is high

Q Find uncertainity in position of electron if uncertainity in velocity is 10^5 m/sec.

Ans $\Delta x \times m\Delta v \geq \frac{h}{4\pi}$

$$\Delta x \times 9.1\times10^{-31}\times10^{5} \geq \frac{6.6\times10^{-34}}{4\times3.14}$$

$$\Delta x = \frac{6.6}{9.1\times4\times3.14}\times10^{-8} \text{ meter}$$

$$= \frac{660}{9.1\times12.5}\times10^{-10}$$

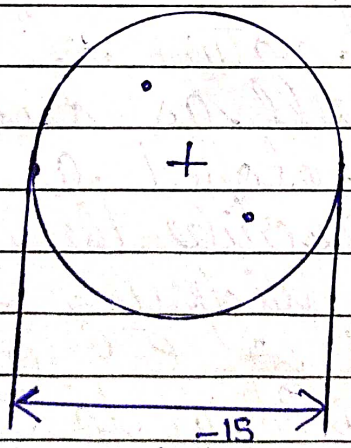
$$= \frac{660}{100}\times10^{-10} \cong 6\times10^{-10} \text{ meter}$$

$$\cong 6\,\text{Å}$$ Ans

Imp

Q Using Heisenberg's uncertainity principle show that e^- doesn't reside in the nucleas.

Ans Let us assume that e^- lives in the nucleas



(In order of) [dia of nucleas]

→ Max. error in position of e^- if inside the nucleas $= 10^{-15}$ m (Δx)

$$\Delta x \times m \Delta v \geq \frac{h}{4\pi}$$

$$10^{-15} \times 9.1 \times 10^{-31} \times \Delta v \geq \frac{6.6 \times 10^{-34}}{4 \times 3.14}$$

$$\Rightarrow \Delta v = \frac{6.6}{110} \times 10^{12}$$

$$= \frac{660}{110} \times 10^{10}$$

$$= 6 \times 10^{10} \text{ m/s}$$

↑ Greater than c

Impossible.

Quantum Mechanical Model of Atom

[Developed by Schrödinger & Heisenberg]

Based on dual nature of electron (particle as well as wave nature), Schrödinger developed a mathematical equation to describe the wave nature of electron considering the bound movement of e^- around the nucleas (similar to Standing Waves)

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

"The Iyrect Schrödinger Wave Equation"

ψ → Amplitude of electron wave or wave function

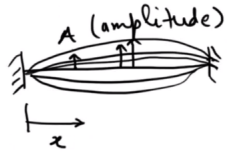
x, y, z → Coordinates

E → Total Energy

V → Potential Energy

m → mass of electron

Standing waves:



$A = f(x)$

↑ amplitude of ✓ standing wave

→ When Schrödinger solved his equation for hydrogen and hydrogen like species for finite, continious and single valued ψ, 3 integers [called Quantum No] are obtained as a natural consequence of solution of Schrödinger wave Eqⁿ.

→ Principal Quantum No. } Obtained from
→ Azimuthal Quantum No. } Schrödinger Wave
→ Magnetic Quantum No. } Eqⁿ

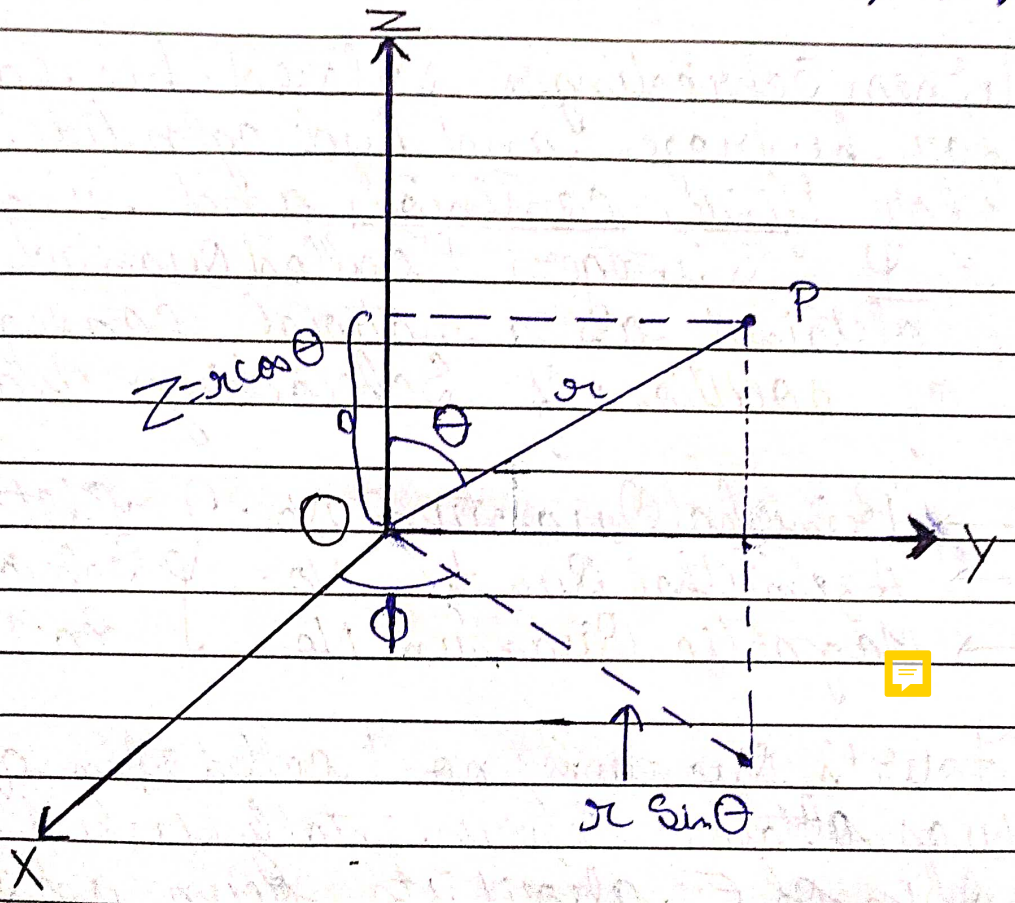
→ Fourth Quantum no. (called Spin quantum no.) was obtained experimentally which is due Spin of e^- about its own axis.

→ An orbital is described by '3' Quantum numbers (Principal, Azimuthal, Magnetic) which were obtained from Schrödinger wave equation.

→ To designate an electron in an atom '4' Quantum no. are needed.
Principal, Azimuthal, Magnetic & Spin Quantum Number.

→ Schrödinger wave equation can be solved more easily in Spherical polar Coordinates. (r, θ, ϕ).

⇒ Casterian Coordinates (x, y, z)
⇒ Spherical Polar Coordinates (r, θ, ϕ)



$\theta \rightarrow$ Polar Angle / colatitude
$\phi \rightarrow$ Azimuthal Angle

$$Z = r\cos\theta$$

$$OP' = r\sin\theta$$

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$x^2 + y^2 + z^2 = r^2$$

→ When Schrödinger eq^n is solved for H atom for ψ, then obtained values of ψ could be factorized into two parts
⇒ One containing only radial distance 'r' and other containing (θ, ϕ)

$$\boxed{\psi(r,\theta,\phi) = \psi_r \times \psi(\theta,\phi)}$$

$\psi(r,\theta,\phi)$ ← Wave function; ψ_r ← Radial Wave function; $\psi(\theta,\phi)$ ← Angular Wave Function

→ $$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

⇒ $$\boxed{\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0}$$

Where ∇^2 → Laplacian Operator $= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

read as 'del squared'

⇒ $$\boxed{\nabla^2\psi \times \frac{h^2}{8\pi^2 m} + (E-V)\psi = 0}$$

⇒ $$\frac{-h^2}{8\pi^2 m}\nabla^2\psi - E\psi + V\psi = 0$$

$$\left(\underbrace{-\frac{h^2}{8\pi^2 m}\nabla^2 + V}_{H}\right)\psi = E\psi$$

⇒ $$H\psi = E\psi$$

where $H = \left(\frac{-h^2}{8\pi^2 m}\nabla^2 + V\right)$

↑ Hamiltonian Operator

Quantum Numbers:

① Principal Quantum Numbers: (n)

→ Proposed by Bohr, it identifies shell.
→ It assumes all the +ve integral values.
⇒ n = 1, 2, 3, 4, ---------

Designation of shell: K, L, M, N ---

→ It determines size and energy of an orbital.

→ No of orbital in n^{th} shell $= n^2$
→ Max no. of electrons in n^{th} shell $= 2n^2$

② Azimuthal Quantum number: (l)
(or Angular Momentum Quantum No.)
(or subsidiary quantum number)

→ Proposed by sommerfeld, it identifies subshell and assumes all the integral values from 0 to (n-1). It depends on 'n'

Values of 'l' = 0, 1, 2, 3, ---

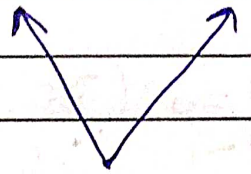
Designation of subshell = s, p, d, f, ---

→ 'l' determines shape and energy of an orbital

* For single e^- species (H, He^+, Li^{2+} etc), energy of an orbital depends only on 'n'

$3s = 3p = 3d$, $4s = 4p = 4d = 4f$

Equal Energy — Equal Energy



For single e^- species.

* However for multi e^- species (He, Na, K etc) energy of orbital depends on both n & l. [n+l rule]

* For multi e^- species

If n is same, then l ↑, energy ↑

$3s < 3p < 3d$

$4s < 4p < 4d < 4f$

increasing energies of orbitals

→ Orbital Angular Momentum $= \sqrt{l(l+1)}\ \frac{h}{2\pi}$

$= \sqrt{l(l+1)}\ \hbar$

Eg For ns orbital, $l = 0$

Orbital angular momentum = 0 for s-orbital

→ No of orbitals in a subshell = $(2l+1)$

→ Max. no of electrons in a subshell = $2(2l+1)$

→ No of subshells in n^{th} shell = n

Eg $n=3$, $l=0$ (3s subshell)
$l=1$ (3p subshell)
$l=2$ (3d subshell)

③ Magnetic Quantum No. (m):

→ Proposed by Linde, it determines orientation of an orbital. It assumes all the integral values from $-l$ to $+l$ including zero

Values of $m = \underbrace{-l, -(l-1), -(l-2), \cdots, -3, -2, -1}_{l \text{ values}}, \underbrace{0}_{\text{one more}}, \underbrace{1, 2, 3, \cdots, (l-1), l}_{l \text{ values}}$

Total no. of 'm' or total no of orientations or total no. of orbitals in a subshell = $(2l+1)$

→ In absence of any external Electric or Magnetic field, various orbitals of a subshell are of equal energy (degenerate orbitals)

$n=1$, $l=0$ (1s subshell), $m=0$ (1s orbital)
$n=2$, $l=0$ (2s subshell), $m=0$ (2s orbital)
1 (2p subshell), $m=-1, 0, +1$ → 3 orbitals
($2p_x$, $2p_y$, $2p_z$) — 4 orbital

Energy are equal in absence of any external Electric or Magnetic field.

$n=3$, $l=0$ (3s subshell), $m=0$ (3s orbitals)
$l=1$ (3p), $m=-1,0,+1$ } $3p_x, 3p_y, 3p_z$
$l=2$ (3d), $m=-2,-1,0,+1,+2$
five degenerate 'd' orbital
($3d_{xy}, 3d_{yz}, 3d_{xz}, 3d_{x^2-y^2}, 3d_{z^2}$)

④ Spin Quantum Number (s)

→ It determines spin of e^- about its own axis.
→ Its two possible values are $+\frac{1}{2}$ and $-\frac{1}{2}$ representing two spin states of e^-.
→ Spin angular momentum of e^-

$$= \sqrt{s(s+1)}\,\frac{h}{2\pi}$$

where $s = \frac{1}{2}$

$$= \sqrt{\frac{1}{2}\times\frac{3}{2}}\times\frac{h}{2\pi}$$

$$= \frac{\sqrt{3}\,h}{4\pi}$$ (a constant value)

→ An orbital can accomodate a maximum of 2 e^-'s of opposite spins.

↑↓ Correct ✓ ↑↑ or ↓↓ Not possible

Electronic Configuration of Atom:

- → Pauli's Exclusion principle
- → Aufbau's Rule
- → Hund's Rule

Pauli's Exclusion Principle:

→ No two electron in an atom can have all four quantum numbers identical.

Or

→ An orbital can accomodate a max of two electrons of opposite spins.

Eg 4S Orbital

For 1st e^- $n=4, l=0, m=0, S=+\frac{1}{2}$

For 2nd e^- $n=4, l=0, m=0, S=-1/2$

For 3rd e^- in "4S" $n=4, l=0, m=0, S=$ either $+\frac{1}{2}$ or $-\frac{1}{2}$

3rd e^- isn't possible in an orbital.

Aufbau's Rule (Madelung rule) :→ This rule is about the relative energies of various orbitals of an atom.

For multi e⁻ species

→ Also known as (n+l) rule.
→ Higher is the (n+l) value of an orbital, higher will be its energy. If two different orbitals in an atom have same (n+l) value, then lower will be the value of n, lower will be energy of orbital.

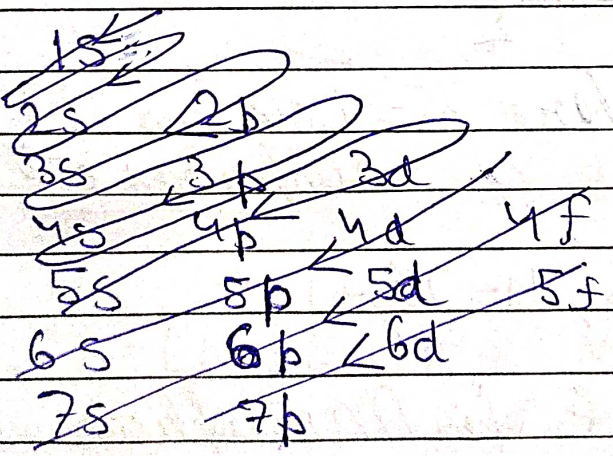
→ eg 2S, 3S, 3d, 4S, 4p, 4d, 4f

(n+l) =	2+0	3+0	3+2	4+0	4+1	4+2	4+3
	=2	=3	=5	=4	=5	=6	=7

2S < 3S < 4S < 3d < 4p < 4d < 4f

Increasing order of energies

⇒



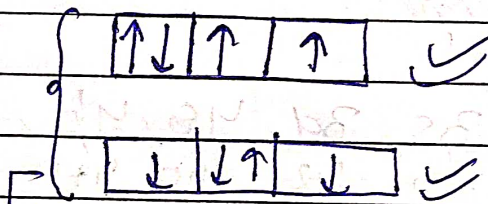
1S < 2S < 2p < 3S < 3p < 4S < 3d < 4p < 5S < 4d < 5p < 6S < 4f < 5d < 6p < 7S - - -

Hund's Rule of Maximum Multiplicity :-

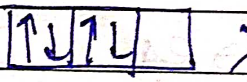
→ This rule is about filling of e^- among degenerate orbitals of a subshell.

→ According to Hund's rule, pairing of e^- among degenerate orbitals of a subshell should occur only when each orbital has got one each of similar spin.

eg $2p^4$:

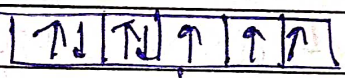


Correct



(Wrong)

eg $3d^7$



3 unpaired electrons

→ Total Spin 'S' $= \frac{n \times 1}{2}$

where n = total no. of unpaired e^-.

→ Spin Multiplicity, $M = 2S + 1$

Q Find total Spin & Spin Multiplicity of
(i) $3p^3$ (ii) $4d^6$

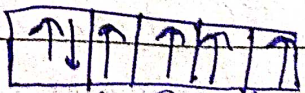
Ans (i) $3p^3 \Rightarrow$

$n = 3$

$S = \frac{1}{2} \times 3$

$M = \frac{2 \times 3}{2} + 1 = 4$

(ii) $4d^6$



4 unpaired e^-

$S = \frac{1}{2} \times 4 = 2$

$M = 2(2) + 1 = 5$

Electronic Configuration of Atom:-

n → Principal Quantum No.

l → Azimuthal No. or Subsidiary Quantum no.

(5) → no of e^- in subshell

eg $2p^5$ ($n=2$, $l=1$)

$_1H$: $1s^1$

$_2He$: $1s^2$, $2s^0$

$_3Li$: $1s^2$, $2s^1$ or $[He]\ 2s^1$ → Condensed E.C

$\Rightarrow [He]\ 2s^1$

↑ Outer Electronic Configuration

$_7N$: $1s^2, 2s^2, 2p^3$ or $[He]\ 2s^2 2p^3$ (Ground State E.C)

or $[He]$ ↑↓ | ↑ | ↑ | ↑ ($n=3$ unpaired e-)

$_{11}Na$: $1s^2, 2s^2, 2p^6, 3s^1$ or $[Ne]\ 3s^1$

$_{13}Al$: $[Ne]\ 3s^2 3p^1$

$_{17}Cl$: $[Ne]\ 3s^2 3p^5$ or $[Ne]$ (3s: ↑↓) (3p: ↑↓ | ↑↓ | ↑) ($n=1$)

(one unpaired e^- in ground state config)

$_{18}Ar$: $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$

$_{19}K$: $[Ar]\ 4s^1$

$_{20}Ca$: $[Ar]\ 4s^2$

3d Series (Sc to Zn)

21 Sc : $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^1$

or $[Ar]\ 4s^2\ 3d^1$ ✓

or $[Ar]\ 3d^1\ 4s^2$ ✓ [Usefull]

22 Ti : $[Ar]\ 3d^2\ 4s^2$

23 V : $[Ar]$ ↑ ↑ ↑ | | (3d) $n=3$

24 Cr* : $[Ar]\ 3d^5\ 4s^1$ ✓

25 Mn : $[Ar]\ 3d^5\ 4s^2$

26 Fe : $[Ar]$ ↑ ↑ ↑ ↑ ↑↓ | ↑↓ $n=4$

27 Co : $[Ar]\ 3d^7\ 4s^2$

28 Ni : $[Ar]\ 3d^8\ 4s^2$

29 Cu* : $[Ar]\ 3d^{10}\ 4s^1$ ✓

30 Zn : $[Ar]\ 3d^{10}\ 4s^2$

36 Kr : $[Ar]\ 3d^{10}\ 4s^2\ 4p^6$

31 Ga : $[Ar]\ 3d^{10}\ 4s^2\ 4p^1$

32 Ge : $[Ar]\ 3d^{10}\ 4s^2\ 4p^2$

33 As : $[Ar]\ 3d^{10}\ 4s^2\ 4p^3$

34 Se : $[Ar]\ 3d^{10}\ 4s^2\ 4p^4$

35 Br : $[Ar]\ 3d^{10}\ 4s^2\ 4p^5$

Exceptional E.C of Cr and Cu :

$_{24}Cr$ →
- Expected : [Ar] $3d^4 4s^2$ ✗
- Correct : [Ar] $3d^5 4s^1$ ✓

$_{29}Cu$ →
- Expected : [Ar] $3d^9 4s^2$ ✗
- Actual : [Ar] $3d^{10} 4s^1$ ✓✓

(I) Symmetrical distribution of electron :

→ Symmetry provides stability

→ Half filled and fully filled electronic configuration in d-subshell (particularly) is more stable configuration due to symmetrical distribution of e^-

$3d^5$ ↑ | ↑ | ↑ | ↑ | ↑

$3d^{10}$ ↑↓ | ↑↓ | ↑↓ | ↑↓ | ↑↓

(II) Exchange Energy :

It is observed that similar spin electrons present in degenerate orbitals of a subshell exchange their positions and release energy.

This stabilising effect is especially more for half filled and fully filled configuration.

Also the released energy due to exchanges is called exchange energy.

→ More the no. of exchanges, greater will be the exchange energy and hence greater stability.

eg

1st 2nd 3rd 4th 5th

↑ | ↑ | ↑ | ↑ | ↑

eg

1st	2nd	3rd	4th	5th
↑	↑	↑	↑	↑

1st e^- = 4
2nd e^- = 3
3rd e^- = 2
4th e^- = 1
Total = 10 exchanges

Total No. of exchange among similiar spin $e^- = {}^nC_2 = \frac{n(n-1)}{2}$

Where n = no. of similiar spin electrons present in degenerate orbitals of a subshell

eg $3d^{10}$,
Total exchanges = 10 + 10
= 20

Electronic Configuration Of Ions: → cation
→ Anion

(a) E.C of cation: → Cations are formed by removing e^- from neutral atom.

→ To form cation, outer e^- are removed from neutral atom. However note that in d-block metals, e^- are first removed from ns orbital and then the penultimate (n-1) d orbital.

Outermost: (n^{th} shell) → (s & p-block)
Penultimate: $(n-1)^{th}$ shell → (d-block)
Antipenultimate: $(n-2)^{th}$ shell → (f-block)

→ eg $_{26}Fe$: [Ar] $3d^6 4s^2$ or [Ar] ↑↓ ↑ ↑ ↑ ↑ ↑ (3d) ↑↓ (4s)

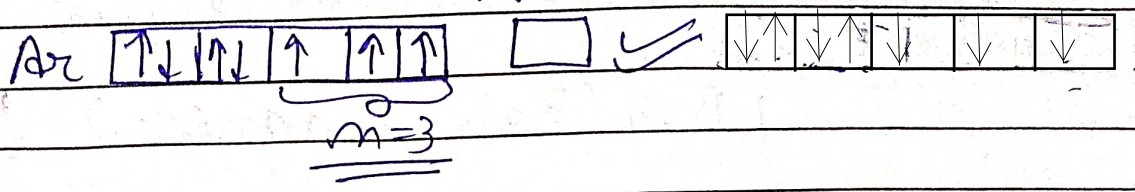
$_{26}Fe^{2+}$: [Ar] $3d^6 4s^0$ or [Ar] ↑↓ ↑ ↑ ↑ ↑ ↑ (3d) □ (4s)

$_{26}Fe^{3+}$: [Ar] $3d^5 4s^0$ or [Ar] ↑ ↑ ↑ ↑ ↑ (3d) □ (4s)
5 unpaired e^-

Note: (III-Adv)

$_{26}Fe$: $[Ar]\,3d^6\,4s^2$

Fe^+: $[Ar]\,3d^6\,4s^1$ converts to $[Ar]\,3d^7\,4s^0$ ✓



(b) Anions: Anions are formed by adding e⁻'s in neutral atom in accordance of the three rules (Pauli, Aufbau, Hund's Rule).

$_9F$: $1s^2, 2s^2 2p^5$

F^-: $1s^2, 2s^2, 2p^6$

Magnetic Nature:

(a) Diamagnetic Nature:-

→ Substance which are weakly repelled under Magnetic field
→ Contain zero unpaired electron
→ Show a slight decrease in weight when placed in magnetic balance.

(b) Paramagnetic Substance:-

→ Substance which are weakly attracted in Magnetic field.
→ contains one or more number of unpaired electrons
→ Show a slight increase in weight when placed in Magnetic Balance.

* Spin magnetic moment $= \sqrt{x(x+2)} \times \left(\frac{eh}{4\pi m_e}\right)$

$= \sqrt{x(x+2)}$ B.M

1 BM (Bohr Magneton) $= \frac{e.h}{4\pi m_e}$

where x = no. of unpaired e⁻.

* For diamagnetic substance, x = 0

Spin magnetic moment = 0

Q Which set of quantum no. is not possible?

(A) n=4, l=1, m=0, s=−1/2

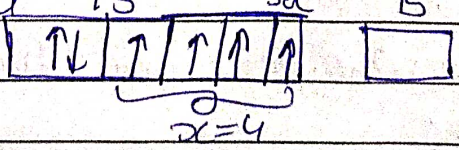
(B) n=5, l=3, m=−3, s=−1/2

(C) n=2, l=2, m=−2, s=+1/2

(D) n=4, l=0, m=0, s=−1/2

Q Find Spin magnetic moment in
(i) Fe^{2+} ion
(ii) Cr^{3+} ion.

Ans (i) Fe^{2+} : [Ar] $3d^6$ $4s^0$



Spin magnetic moment $= \sqrt{4 \times 6}$ BM

$= \sqrt{24}$ BM

(ii) Cr^{3+} : $[Ar]\ 3d^3\ 4s^0$

or $[Ar]$ ☐ ☐ ↑ ↑ ↑ ☐

$x = 3$

Spin magnetic moment $= \sqrt{3 \times 5} = \sqrt{15}\ BM$

IIT-Adv (2015)

Q (Imp) Find no of degeneracies in 2nd excited state of H^- ion.

Ans

ground state

H: 1S, 2S = 2p, 3S = 3p = 3d

9 degeneracies.

1st excited state, 2nd excited state

H^- : (2e)

1st

1S, 2S, 2p → 3 degeneracies in 2nd ES of ~~multi e~~ H^-.

1S: ground; 2p: 2nd ~~excited state~~

Ans : 3

Node / Nodal Plane :

Schrodinger wave equation

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

→ ψ : Wave function (or Amplitude of e wave)

→ $\psi(r, \theta, \phi) = \psi_r \times \psi(\theta, \phi)$

$\psi(r, \theta, \phi)$: Wave function; ψ_r: Radial wave function; $\psi(\theta, \phi)$: Angular wave fxn

Physical Significance of ψ : ψ (wave function) is the amplitude of wave nature of e^-. It can be +ve or -ve depending upon coordinates. As such there is no physical significance of ψ.

Physical Significance of ψ^2 :

→ As per classical theory of wave motion (Maxwell's wave theory), intensity of wave is directly proportional to the square of amplitude.

→ Max Born suggested that square of wave function ψ should represent the probability of finding e^- per unit volume or probability density.

⇒ ψ^2 = Probability density i.e probability of finding e^- per unit volume at a point around nucleus.

→ $\psi(r,\theta,\phi) = \psi_r \times \psi(\theta,\phi)$

$\psi^2(r,\theta,\phi) = \psi_r^2 \times \psi^2(\theta,\phi)$

↑ Probability density; ↑ Radial Probability density; ↖ Angular Probability Density.

→ Probability of finding e^- in a volume dV around nucleus = $\psi^2 \times dV$

→ **Node:** It is a plane or surface where probability of finding e^- is zero.

⇒ At node : $\psi^2 = 0 \Rightarrow \psi = 0$ (Imp)

Node

- Radial Node or Spherical Node
- Angular node

(a) **Radial node (or Spherical Node):** It is a spherical shell at a finite distance from nucleus where radial Probability of finding e^- is zero.

$\psi_r^2 = 0$ for radial node

No of radial node in an orbital = $n-l-1$

(b) **Angular Node:** It is a plane or surface passing through nucleus where probability of finding e^- is zero.

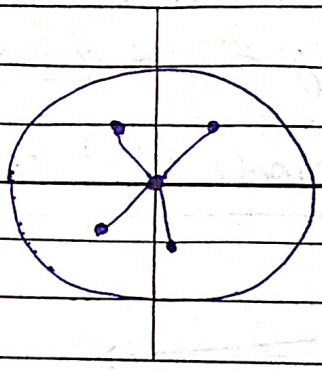
$\psi^2_{(\theta,\phi)} = 0$ for angular node.

No of angular node in an orbital = l

⇒ Total no. of nodes in an orbital = $(n-l-1)+l$
$= n-1$

Shape Of Orbitals :

S-orbital : Spherical (non-directional)



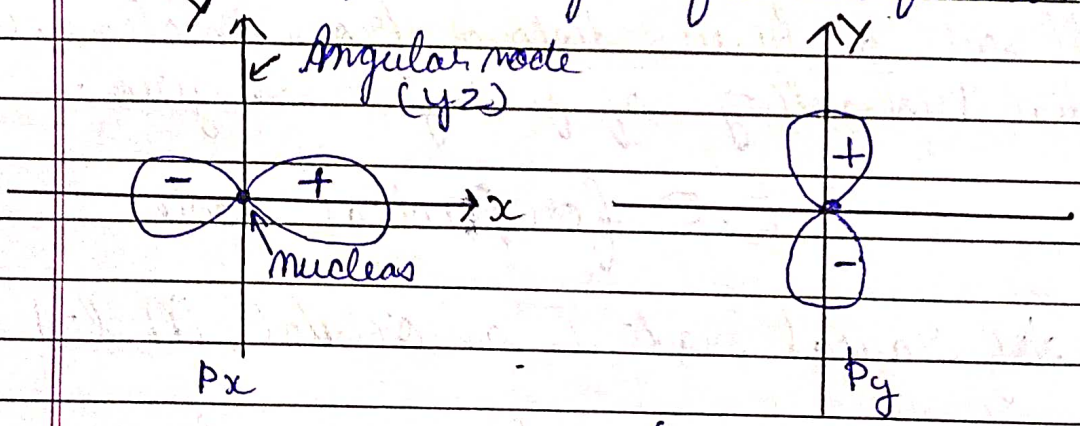
No. of angular node in 'S' orbital = 0.

S-orbital

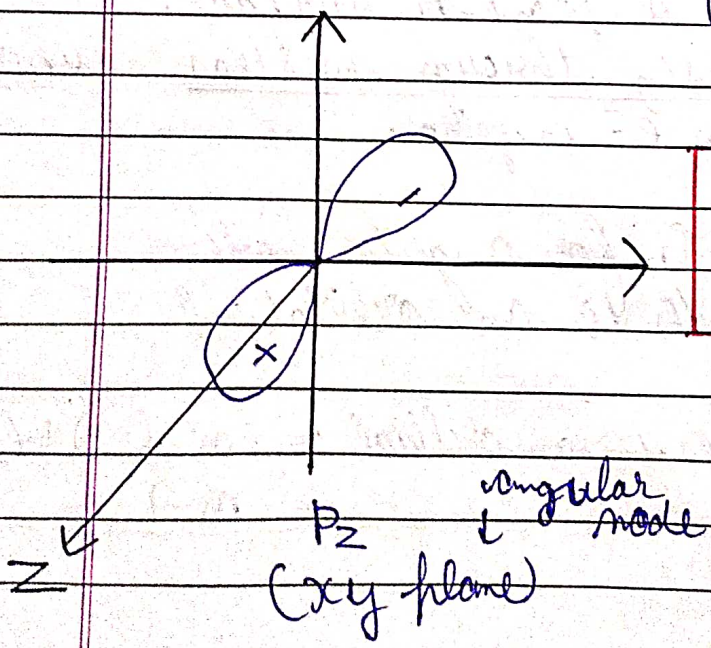
p-orbital : Dumbell Shaped (directional)

No of angular node in each 'p' orbital = 1

⇒ +, − represent sign of wave function (Ψ)



(xz is angular node)

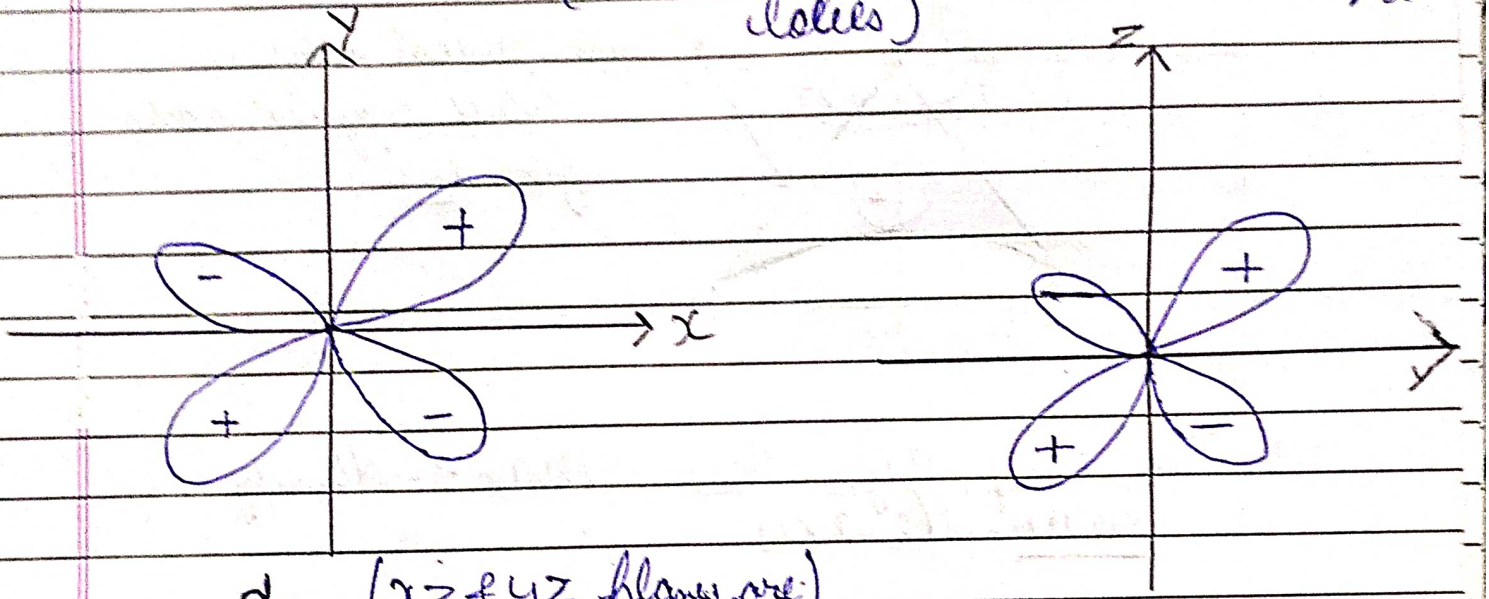


⇒ In p-orbital, maximum e⁻ density lies along the axes

d - Orbitals : ($d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}$, d_{z^2})
(all directional)

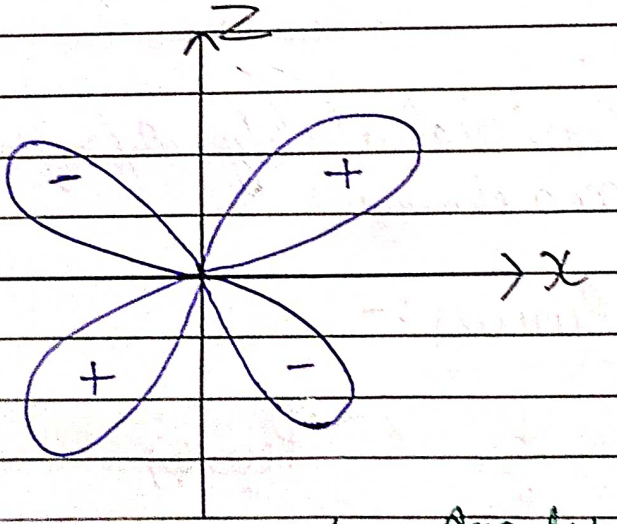
double dumbell shape (each orbital contains 4 lobes)

Donut shaped

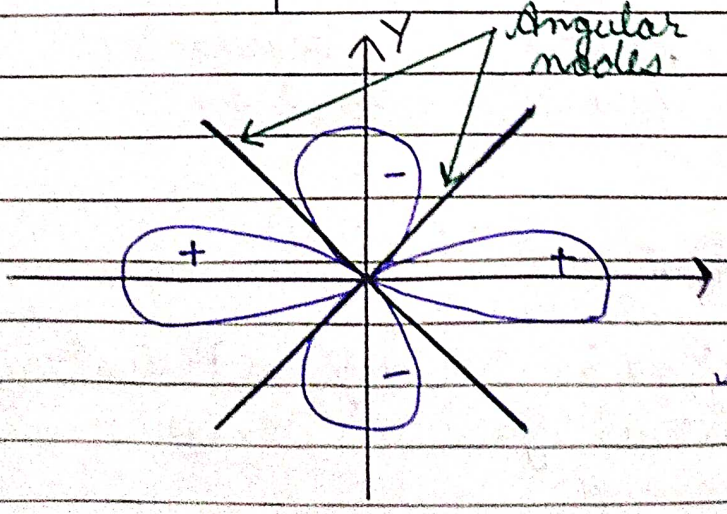


d_{xy} (xz & yz planes are angular nodes)

d_{yz} (xy & xz plane)

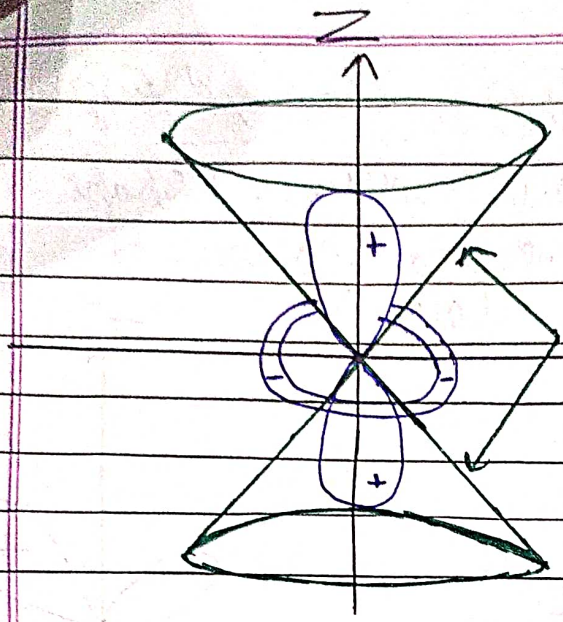


d_{xz} (xy & yz are angular nodes)



$d_{x^2-y^2}$ ($Y = |x|$ with z-axis)

* In $d_{x^2-y^2}$, angular node are at 45° from x & y axis.



d_{z^2}

Two conical surfaces are angular nodes for d_{z^2}

→ In d_{xy}, d_{yz}, d_{zx}, Max e^- density lies between the axes

→ In $d_{x^2-y^2}$, d_{z^2}, max e^- density lies along the axes.

⇒ Antinode: Where Probability of finding e^- is maximum.

Probability Curves :-

$\Psi_{(r,\theta,\phi)} = \Psi_r \times \Psi_{(\theta,\phi)}$

↑ Wave fsn (Amplitude of e^- wave) ↑ Radial Wave fsn ↑ Angular wave fsn

$\Psi^2_{(r,\theta,\phi)} = \Psi^2_r \times \Psi^2_{(\theta,\phi)}$

↑ Probability Density ↑ Radial Probability Density ↑ Angular Probability Density

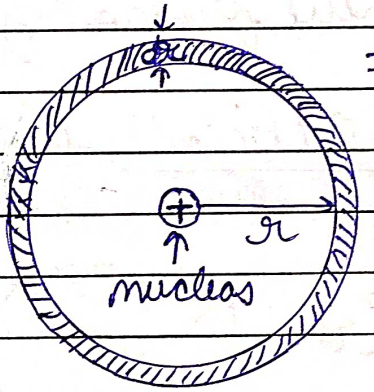
Radial Probability Distribution function, $4\pi r^2 \psi_r^2$

ψ_r^2 → Radial probability density i.e radial probability of finding e^- per unit volume.

Radial Probability of finding $e^- = \psi_r^2 \times dV$

→ Radial Probability of finding e^- in a shell of thickness dr at a radial distance r

$= \psi_r^2 \times V_{shell}$



$\Rightarrow V_{shell} = \frac{4\pi}{3}(r+dr)^3 - \frac{4\pi r^3}{3}$

$= \frac{4\pi}{3}[dr^3 + 3r^2 dr + 3r(dr)^2]$

$= \frac{4\pi \times 3}{3} r^2 dr$

$V_{shell} = 4\pi r^2 dr$

→ Radial Probability of finding e^- in a shell of thickness dr at a radial distance 'r'.

$= \psi_r^2 \times 4\pi r^2 dr$

$= \boxed{4\pi r^2 \psi_r^2 \, dr}$

Radial distribution function
(or Radial Probability distribution function)

⇒ Total radial probability of finding e^- in a sphere of radius r $= \int_{r=0}^{r} 4\pi r^2 \psi_r^2 \, dr$

⇒ To draw ψ_r vs r, ψ_r^2 vs r, $4\pi r^2 \psi_r^2$ vs r for various orbitals

(V. Imp)

⇒ **In general :**

$\psi_r = K r^l \times$ (a polynomial in r with deg. $(n-l-1)$) $\times e^{-r/K'}$

↑

Radial wave function for any function

eg is **orbital**: $l=0$, radial node $= 1-0-1 = 0$

$$\psi_r = K e^{-r/K'}$$

⇒ 1S : $\psi_r = K e^{-r/K'}$

2S : $\psi_r = K(K'-r) e^{-r/K''}$

3S : $\psi_r = K(K'-K''r+r^2) e^{-r/K''}$

2p : $\psi_r = Kr e^{-r/K'}$

3p : $\psi_r = Kr(K'-r) e^{-r/K''}$

4d : $\psi_r = Kr^2(K'-r) e^{-r/K''}$

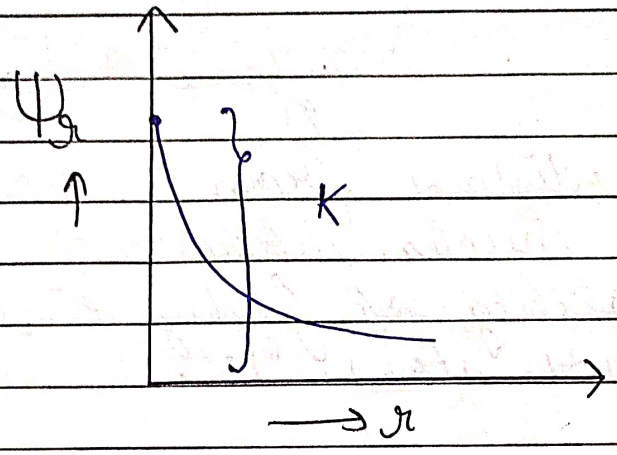
⇒ Graphs for 1S : $\psi_r = Ke^{-Br/K'}$

*In fact for H-atom $K' = a_0$ = Bohr Radius

$$= \frac{h^2}{4\pi^2 m_0 e^2}$$

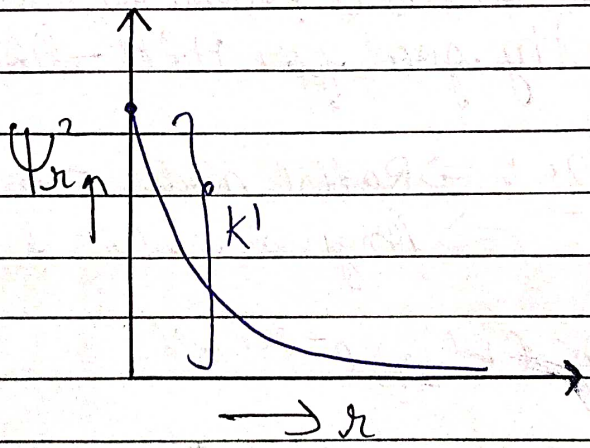
$$= 0.529\ \text{Å}$$

$$\psi_r = K(e^{-r/a_0})$$



ψ_r VS r

$$\psi_r^2 = K'(e^{-2r/a_0})$$

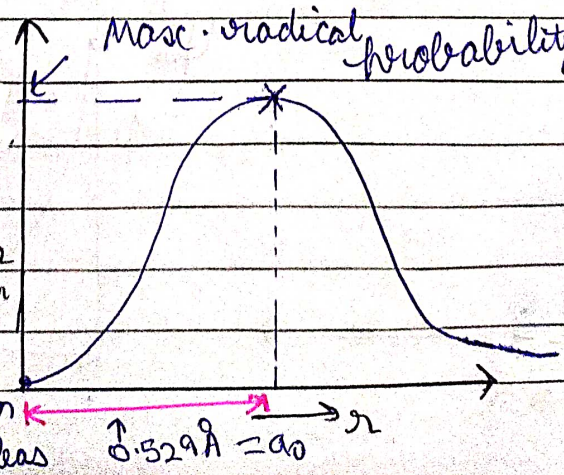


ψ_r^2 VS r

$$y = 4\pi r^2 \psi_r^2 = Kr^2 e^{-2r/a_0}$$

r^2 → Pre-expo factor

$y = 4\pi r^2 \psi_r^2$



$4\pi r^2 \psi_r^2$ VS r

↑ Radial Distribution function

For H atom (1S)

For max radial probability $\frac{d}{dr}(4\pi r^2 \psi_r^2) = 0$

$$\Rightarrow \frac{d}{dr}\left(K r^2 e^{-2r/a_0}\right) = 0$$

$$r^2 \times e^{-2r/a_0} \times \left(\frac{-2}{a_0}\right) + e^{-2r/a_0} \times 2r = 0$$

$$\frac{-2r}{a_0} = -2$$

$$\Rightarrow r_{max} = a_0 = \frac{h^2}{4\pi^2 m K e^2}$$

r → distance from nucleus where probability of finding e^- is max for H atom in 1s.

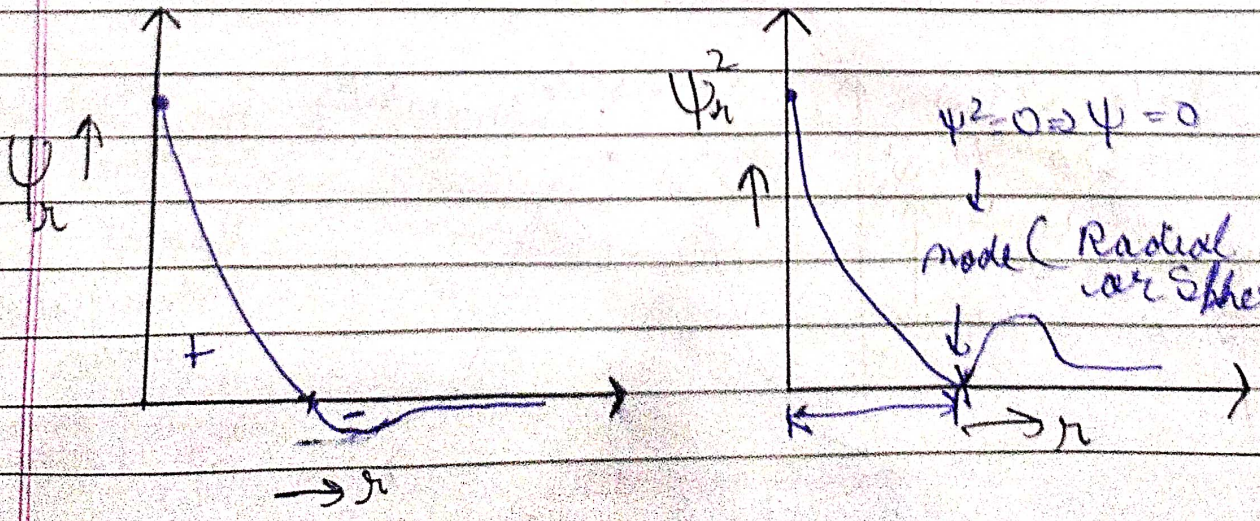
$= 0.529 \text{ Å}$

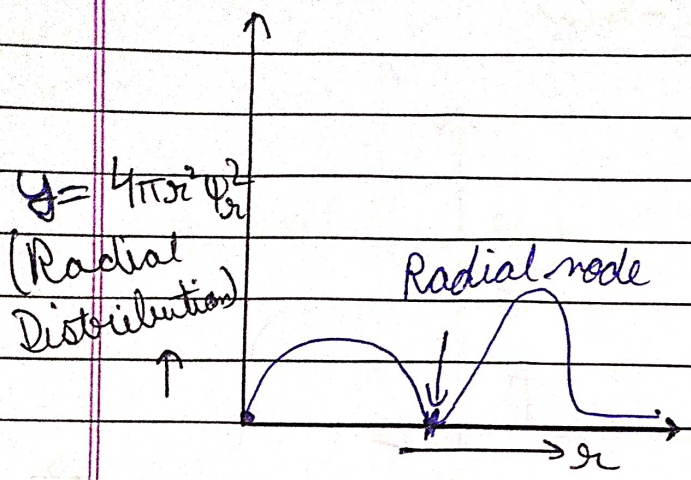
$\Rightarrow$ This implies that results of Bohr model for H and outcome of Quantum Mechanical Model are same $\Rightarrow$ Bohr model was pretty good for H & H-like species

Graphs for 2s: $\Rightarrow$ Radial nodes $= n - l - 1 = 2 - 0 - 1 = 1$

$\Rightarrow$ Angular nodes $= l = 0$

$$(\psi_r)_{2s} = K(K' - r)e^{-r/K''}$$

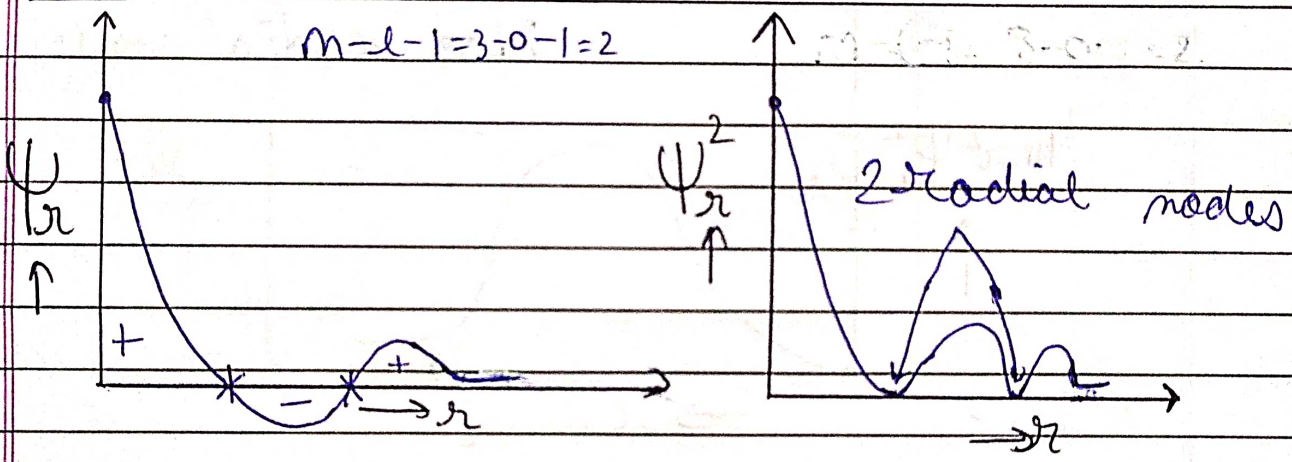


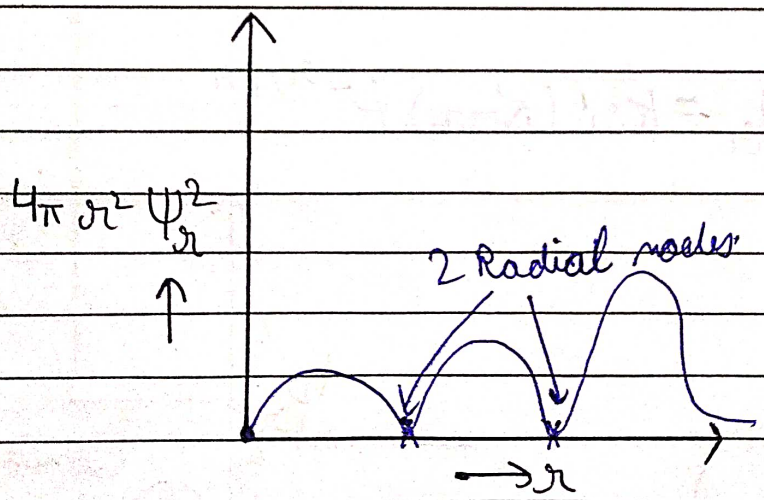


$y = K r^2 (K' - r^*)^2 e^{-2r/K''}$

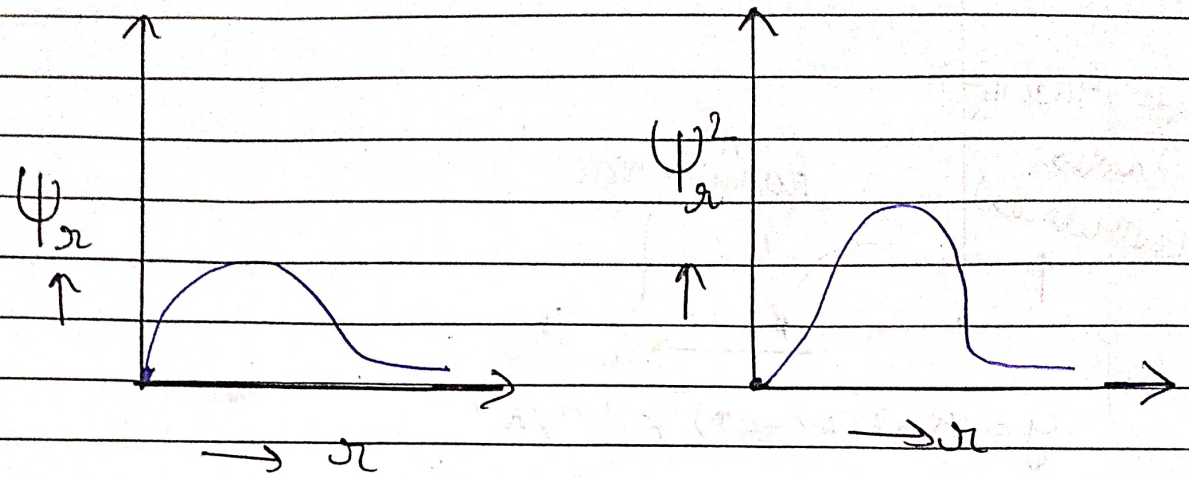
3S

$n - l - 1 = 3 - 0 - 1 = 2$

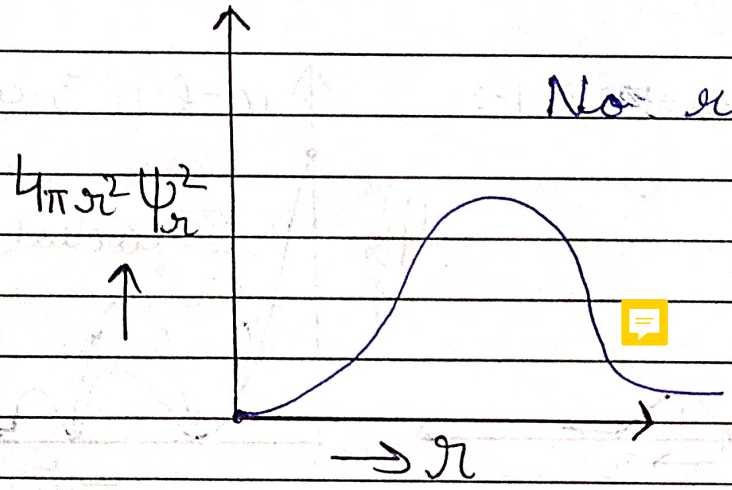




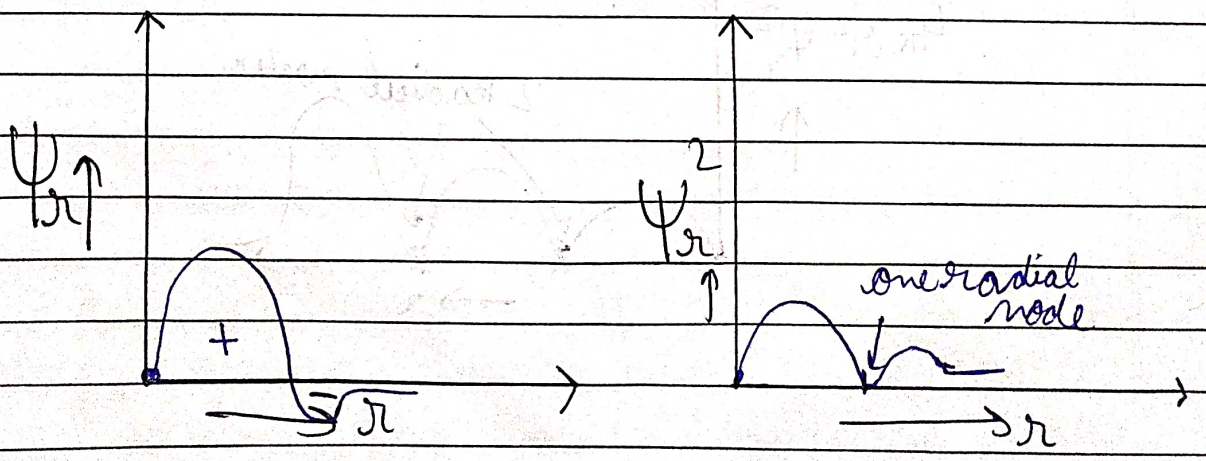
2p $\psi_r = K r' e^{-r/K'}$

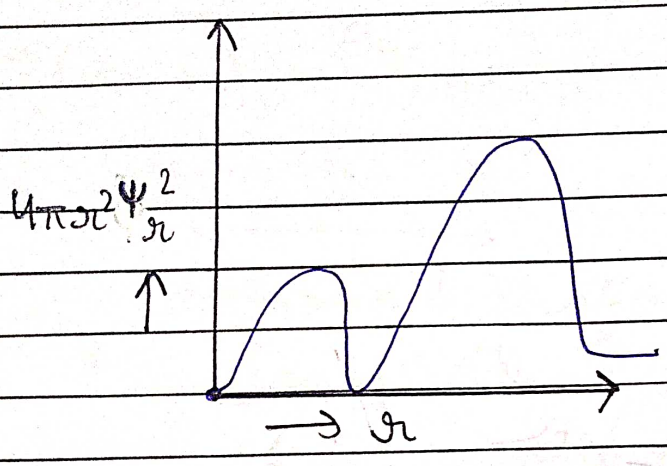


No. radial node

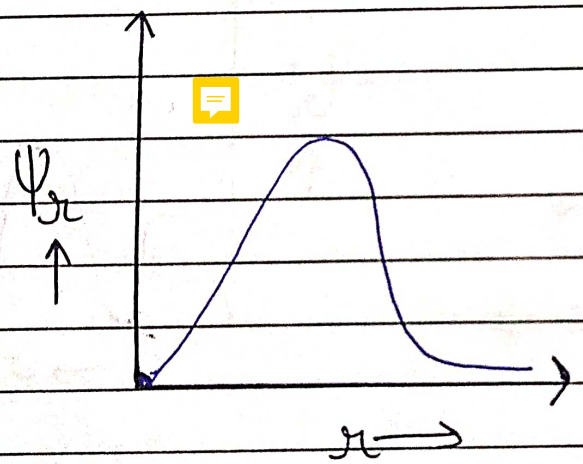


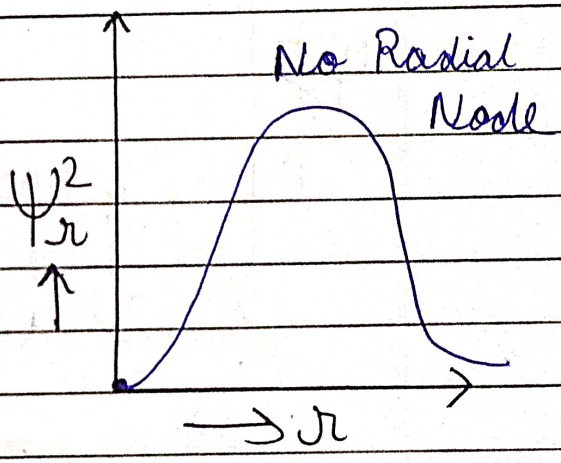
3p $\psi_r = K r' (K' - r_1) e^{-r/K''}$

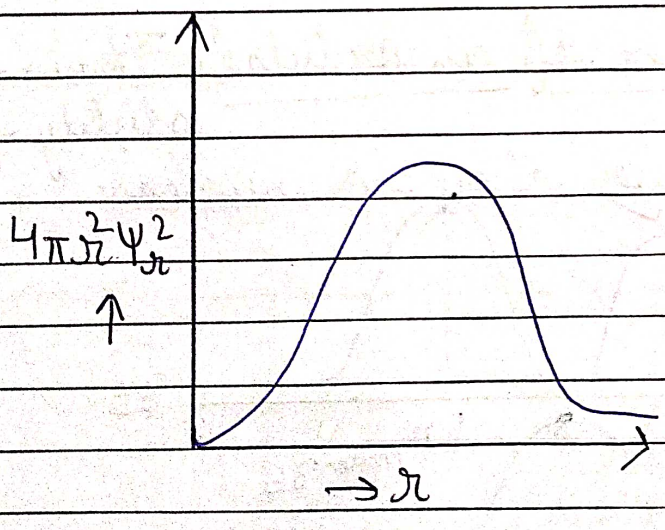




3d $\psi_r = K r^2 e^{-r/K1}$

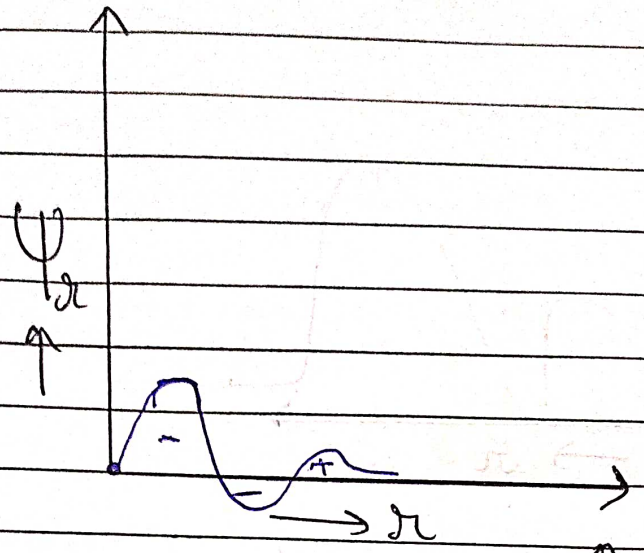


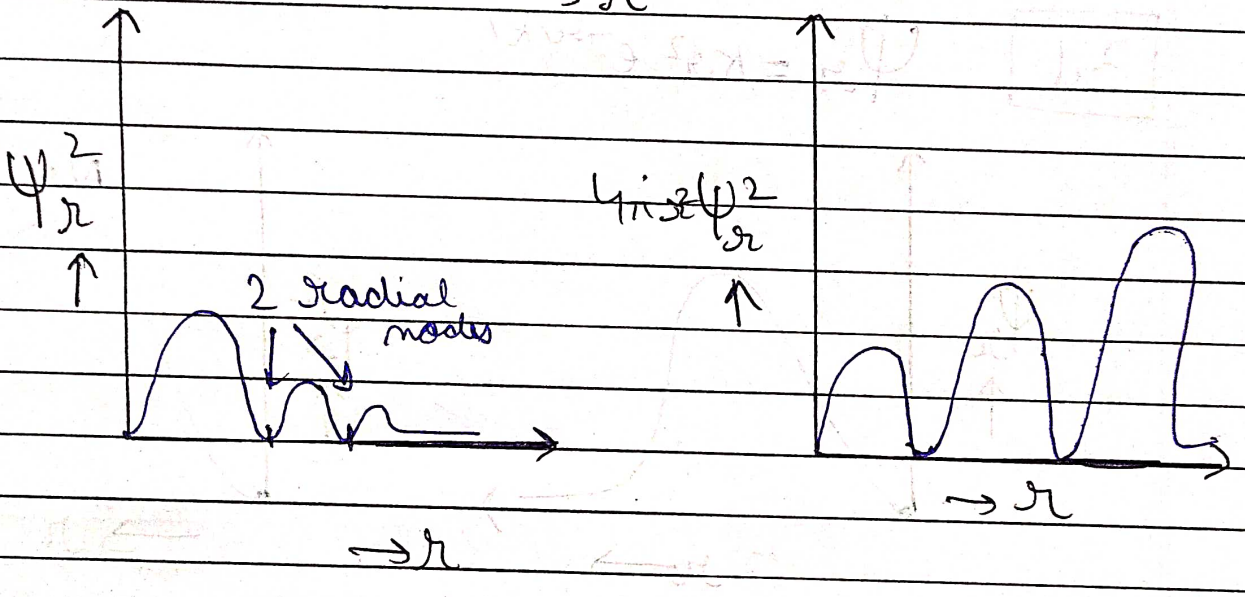




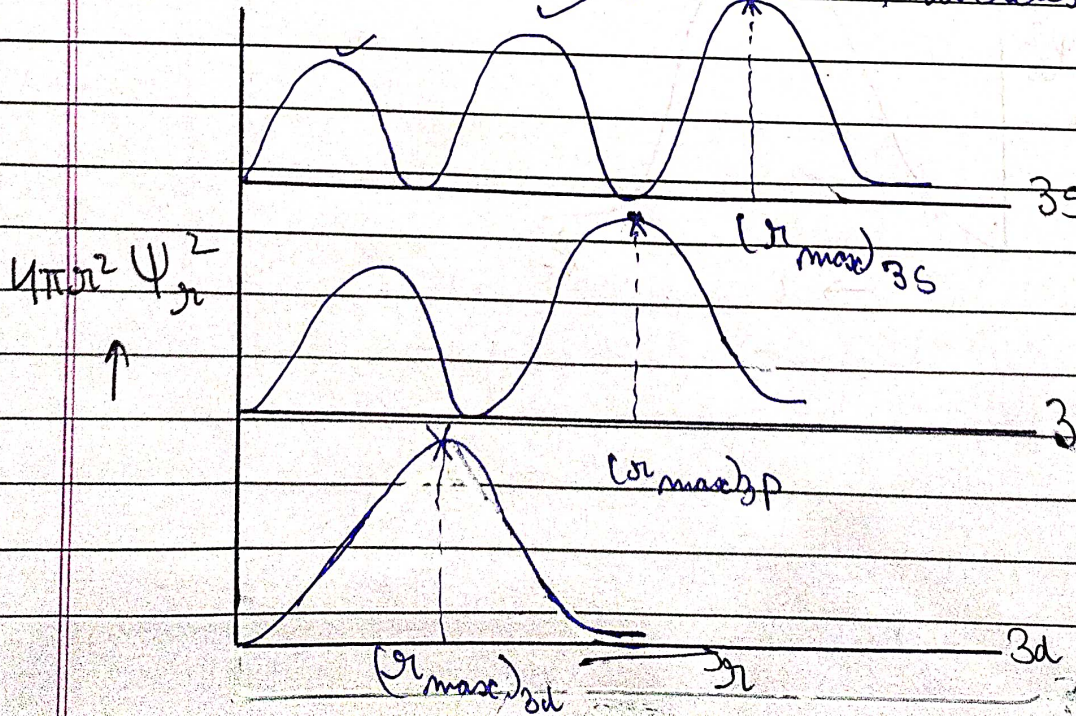
Sol:

$n-l-1$
$= 5-2-1$
$= 2$





⇒ Penetration Power of an orbital: It is the ability of an orbital to be close to nucleus.



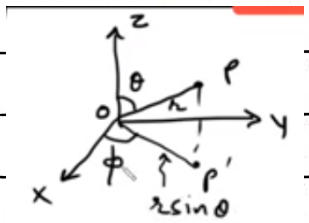
→ $(r_{max})_{3d} < (r_{max})_{3p} < (r_{max})_{3s}$

→ $3s > 3p > 3d$

decreasing Penetration power

* 3s has greater penetration power due to two additional peaks signalling that e^- spend more time close to nucleus.

× × ×



$\Rightarrow \psi_{(r,\theta,\phi)} = \psi_r \times \psi_{(\theta,\phi)}$

Imp $\psi_{(\theta,\phi)}$ = Constant for ns orbital (do not depend on θ, ϕ)

$\psi_{(\theta,\phi)} = f(\cos\theta)$ for np_z orbital

$\psi_{(\theta,\phi)} = f(\sin\theta\cos\phi)$ for np_x orbital

$\psi_{(\theta,\phi)} = f(\sin\theta\sin\phi)$ for np_y orbital

× × ×

The End